

ECE 332: Engineering Electromagnetics II

Lecture 19

Mark A. Martin

Portland State University

June 4, 2026

Exam 2 Logistics

- Unless you have an accommodation, the exam will be held **in person, on paper** in **FAB 48** on Tuesday, June 9, from 1:30 – 3:20 pm.
- If you have an **accommodation**, I will contact you separately to discuss details. If you follow the "Testing Services" link on the class Canvas site, you should be able to arrange to take the exam with your accommodation at the Testing Center.
- The exam will cover the material from **Chapters 8 & 9** of the text.

Main topics:

- Reflection and Transmission at a Boundary between Materials with Different EM Properties
 - normal incidence in lossless and lossy media
 - geometric optics model – Snell's laws, critical angle and total internal reflection, fiber optics
 - general model of oblique incidence using polarization relative to the plane of incidence – Brewster angle, reflectivity and transmissivity

Exam 2 Logistics (Continued)

- Waveguides and Cavities
- Radiation and Antennas
 - quantities that characterize antennas
 - effective area of a receiving antenna
 - current-source antennas – dipole antennas
 - aperture-field antennas – large aperture antennas
 - antenna arrays – linear arrays, electronic steering/scanning
- **I will provide space for your answers on the exam papers.** But, if you're worried about running out, feel free to bring some extra sheets of paper and a stapler (or other way to fasten papers together).

Exam 2 Logistics (Continued)

- The problems must **mostly be done by hand**.
 - A **calculator that is approved for taking the FE exam** ([list of approved calculators](#)) is allowed for arithmetic and evaluation of functions, such as trigonometric functions, exponential functions, and logarithms.
 - **Calculus must be done by hand.**
 - The **textbook or any other electronic devices will not be allowed.**
 - **I will provide any needed tables.**
- You'll be allowed a **two-sided, 1-page summary of important facts and formulas** to use during the exam. Aside from being a useful reference, the process of creating a summary can be helpful.
I will not collect these.
- Please **show all work**. I can't give credit for work you don't show.

Exam 2 Logistics (Continued)

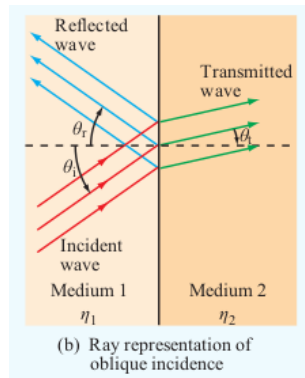
- **I will be in the room to answer any questions you might have during the exam.** Please come up to the lectern if you have a question. I will share any answers that could be beneficial with the rest of the class.
- I will help keep track of time by **displaying a clock** on the screen and **announcing a few time points**.
- **Comments/scores will be written on your exam papers.** Overall scores will be **posted in Canvas**. I will make an announcement in Canvas when the scores are available and provide some statistics.
- **Please remember to take slow, deep breaths and remember that there isn't enough time for me to ask you complicated questions.**

Wave Reflection and Transmission at a Boundary

Chapter 8: Wave Reflection and Transmission at a Boundary

Suppose Medium 1 and Medium 2, with different electromagnetic properties, meet and that an EM wave traveling in Medium 1 reaches the boundary. Then

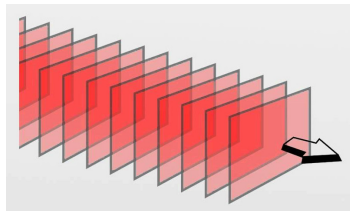
- A portion of the wave is **transmitted** into Medium 2.
- Another portion of the wave is **reflected** back into Medium 1.
- The **direction of propagation of the transmitted wave is different** from the original direction of propagation.
- The **speed of propagation changes** in Medium 2.



Review for Exam 2 (Continued)

Some useful terminology:

- A **ray** is a line in the direction $\hat{\mathbf{k}}$ that an EM wave travels.
- A **wavefront** is a surface where the phase of a wave is constant.
- The wave traveling toward the boundary is the **incident wave**.
- The wave transmitted through the boundary is the **transmitted wave**.
- The wave reflected back into Medium 1 is the **reflected wave**.



Reflection and Transmission at Normal Incidence

- If **direction of travel** of the incident wave is **normal to the boundary**, then the reflected and transmitted waves travel normal to the boundary.
- This is analogous to what occurs at the boundary between two **transmission lines**.

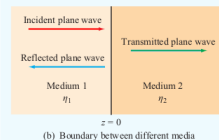
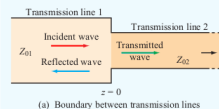
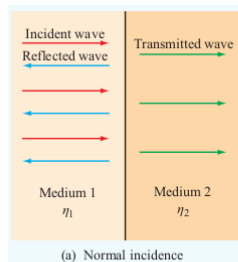


Figure 8-2 Discontinuity between two different transmission lines is analogous to that between two dissimilar media.

Reflection and Transmission at Normal Incidence (Cont.)

Suppose both media are **lossless** and that the **incident wave travels in the direction of the normal to the boundary**.

Medium 1

- Fills half-space $z \leq 0$
- Permittivity ϵ_1
- Permeability μ_1
- $k_1 = \omega\sqrt{\mu_1\epsilon_1}$, $\eta_1 = \sqrt{\mu_1/\epsilon_1}$

Medium 2

- Fills half-space $z \geq 0$
- Permittivity ϵ_2
- Permeability μ_2
- $k_2 = \omega\sqrt{\mu_2\epsilon_2}$, $\eta_2 = \sqrt{\mu_2/\epsilon_2}$

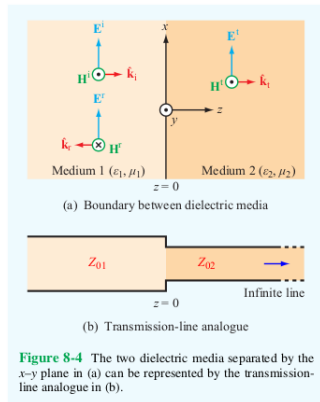


Figure 8-4 The two dielectric media separated by the x - y plane in (a) can be represented by the transmission-line analogue in (b).

Incident wave: x -polarized and traveling in direction $\hat{\mathbf{k}}_i = \hat{\mathbf{z}}$.

Reflection and Transmission at Normal Incidence (Cont.)

Incident Wave ($\mathbf{E}^i, \mathbf{H}^i$)

$$\tilde{\mathbf{E}}^i(z) = E_0^i e^{-jk_1 z} \hat{\mathbf{x}} \quad (1)$$

$$\tilde{\mathbf{H}}^i(z) = (E_0^i / \eta_1) e^{-jk_1 z} \hat{\mathbf{y}} \quad (2)$$

Reflected Wave ($\mathbf{E}^r, \mathbf{H}^r$)

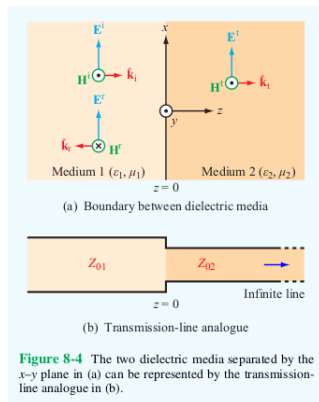
$$\tilde{\mathbf{E}}^r(z) = E_0^r e^{jk_1 z} \hat{\mathbf{x}} \quad (3)$$

$$\tilde{\mathbf{H}}^r(z) = -(E_0^r / \eta_1) e^{jk_1 z} \hat{\mathbf{y}} \quad (4)$$

Transmitted Wave ($\mathbf{E}^t, \mathbf{H}^t$)

$$\tilde{\mathbf{E}}^t(z) = E_0^t e^{-jk_2 z} \hat{\mathbf{x}} \quad (5)$$

$$\tilde{\mathbf{H}}^t(z) = (E_0^t / \eta_2) e^{-jk_2 z} \hat{\mathbf{y}} \quad (6)$$



E_0^i , E_0^r , and E_0^t are the **amplitudes** of $\mathbf{E}^i$, $\mathbf{E}^r$, and $\mathbf{E}^t$ at $z = 0$, resp.

Reflection and Transmission at Normal Incidence (Cont.)

- The **tangential boundary conditions** imply

$$E_0^r = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right) E_0^i = \Gamma E_0^i, \quad E_0^t = \left(\frac{2\eta_2}{\eta_2 + \eta_1} \right) E_0^i = \tau E_0^i, \quad (7)$$

where the **reflection coefficient** is

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (8)$$

and the **transmission coefficient** is

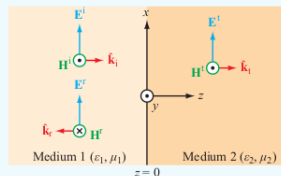
$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1} = 1 + \Gamma. \quad (9)$$

- When the media are **lossless**, Γ and τ are **real-valued**.
- Nonmagnetic media:** $\mu_1 = \mu_2 = \mu_0$

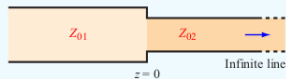
$$\eta_1 = \eta_0 / \sqrt{\epsilon_{r1}}, \quad \eta_2 = \eta_0 / \sqrt{\epsilon_{r2}}, \quad \text{and} \quad \Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}}. \quad (10)$$

Reflection and Transmission at Normal Incidence (Cont.)

- The case of two transmission lines – a lossless line (Line 1) and an infinitely-long lossless line (Line 2) that meet at $z = 0$, is analogous to the case of a normally incident EM wave discussed earlier.
- The transmission-line quantities $(\tilde{V}, \tilde{I}, \beta, Z_0)$ and plane-wave quantities $(\tilde{E}, \tilde{H}, k, \eta)$ **satisfy the same equations**, respectively.



(a) Boundary between dielectric media



(b) Transmission-line analogue

Figure 8-4 The two dielectric media separated by the x - y plane in (a) can be represented by the transmission-line analogue in (b).

Reflection and Transmission at Normal Incidence (Cont.)

- Lessons learned for transmission lines in Chapter 2 can be directly applied to normally-incident EM waves.

Table 8-1 Analogy between plane-wave equations for normal incidence and transmission-line equations, both under lossless conditions.

Plane Wave [Fig. 8-4(a)]	Transmission Line [Fig. 8-4(b)]
$\tilde{\mathbf{E}}_1(z) = \hat{x}E_0^i(e^{-jk_1z} + \Gamma e^{jk_1z})$ (8.11a)	$\tilde{V}_1(z) = V_0^+(e^{-j\beta_1z} + \Gamma e^{j\beta_1z})$ (8.11b)
$\tilde{\mathbf{H}}_1(z) = \hat{y}\frac{E_0^i}{\eta_1}(e^{-jk_1z} - \Gamma e^{jk_1z})$ (8.12a)	$\tilde{I}_1(z) = \frac{V_0^+}{Z_{01}}(e^{-j\beta_1z} - \Gamma e^{j\beta_1z})$ (8.12b)
$\tilde{\mathbf{E}}_2(z) = \hat{x}\tau E_0^i e^{-jk_2z}$ (8.13a)	$\tilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2z}$ (8.13b)
$\tilde{\mathbf{H}}_2(z) = \hat{y}\tau \frac{E_0^i}{\eta_2} e^{-jk_2z}$ (8.14a)	$\tilde{I}_2(z) = \tau \frac{V_0^+}{Z_{02}} e^{-j\beta_2z}$ (8.14b)
$\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$	$\Gamma = (Z_{02} - Z_{01})/(Z_{02} + Z_{01})$
$\tau = 1 + \Gamma$	$\tau = 1 + \Gamma$
$k_1 = \omega\sqrt{\mu_1\epsilon_1}, \quad k_2 = \omega\sqrt{\mu_2\epsilon_2}$	$\beta_1 = \omega\sqrt{\mu_1\epsilon_1}, \quad \beta_2 = \omega\sqrt{\mu_2\epsilon_2}$
$\eta_1 = \sqrt{\mu_1/\epsilon_1}, \quad \eta_2 = \sqrt{\mu_2/\epsilon_2}$	Z_{01} and Z_{02} depend on transmission-line parameters

Reflection and Transmission at Normal Incidence (Cont.)

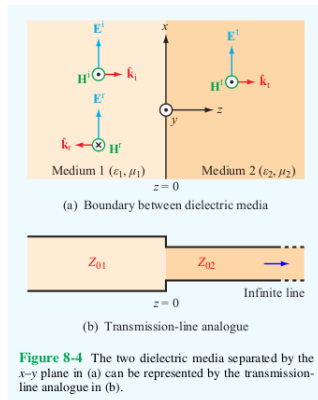
- If the characteristic impedances of Line 1 and Line 2 are Z_{01} and Z_{02} , resp., the **voltage reflection coefficient** at $z = 0$ from Line 1 to Line 2

$$\Gamma = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}} \quad (11)$$

is analogous to the **reflection coefficient** Γ for the EM wave.

- The incident and reflected waves create a **standing wave pattern** in Medium 1, described by the **standing-wave ratio**

$$S = \frac{|\tilde{E}_1|_{\max}}{|\tilde{E}_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \quad (12)$$



Reflection and Transmission at Normal Incidence (Cont.)

- The **locations** $l_{\max}$ where the **electric field intensity reaches its maximum value** in Medium 1 are

$$l_{\max} = -z = \frac{\theta_{\Gamma} + 2n\pi}{2k_1} = \frac{\theta_{\Gamma}\lambda_1}{4\pi} + \frac{n\lambda_1}{2}, \quad (13)$$

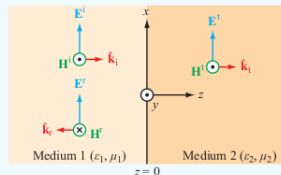
where

$$\Gamma = |\Gamma|e^{j\theta_{\Gamma}} \text{ with } \theta_{\Gamma} \in (-\pi, \pi], \quad (14)$$

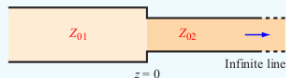
$$\lambda_1 = 2\pi/k_1, \quad (15)$$

and

$$n = \begin{cases} 1, 2, \dots & \theta_{\Gamma} < 0 \\ 0, 1, 2, \dots & \theta_{\Gamma} \geq 0. \end{cases} \quad (16)$$



(a) Boundary between dielectric media



(b) Transmission-line analogue

Figure 8-4 The two dielectric media separated by the x - y plane in (a) can be represented by the transmission-line analogue in (b).

Reflection and Transmission at Normal Incidence (Cont.)

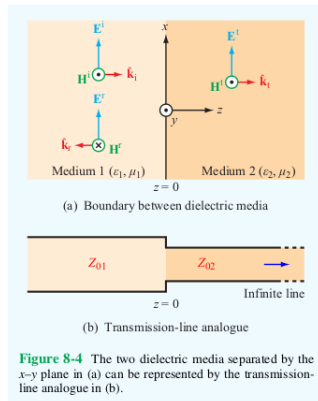
- The **locations** $l_{\min}$ where the **electric field intensity reaches its minimum value** in Medium 1 are

$$l_{\min} = \begin{cases} l_{\max} + \lambda_1/4, & l_{\max} < \lambda_1/4 \\ l_{\max} - \lambda_1/4, & l_{\max} \geq \lambda_1/4. \end{cases} \quad (17)$$

- When the media are **lossless**,

$$\theta_{\Gamma} = \begin{cases} 0, & \eta_2 > \eta_1 \\ \pi, & \eta_2 < \eta_1. \end{cases} \quad (18)$$

- The expressions for $l_{\max}$ and $l_{\min}$ are also valid when Medium 1 is a **low-loss dielectric** and Medium 2 is a **dielectric or conductor**.



Reflection and Transmission at Normal Incidence (Cont.)

Power Flow in Lossless Media

As before, an **x-polarized EM wave** travels in the $+z$ direction **normal** to the **boundary between two lossless media** and intersects the boundary, which is located at the x - y plane.

Incident Wave ($\mathbf{E}^i, \mathbf{H}^i$)

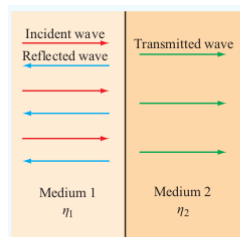
$$\tilde{\mathbf{E}}^i(z) = E_0^i e^{-jk_1 z} \hat{\mathbf{x}} \quad (19)$$

$$\tilde{\mathbf{H}}^i(z) = (E_0^i / \eta_1) e^{-jk_1 z} \hat{\mathbf{y}} \quad (20)$$

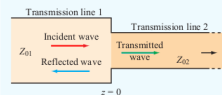
Reflected Wave ($\mathbf{E}^r, \mathbf{H}^r$)

$$\tilde{\mathbf{E}}^r(z) = \Gamma E_0^i e^{jk_1 z} \hat{\mathbf{x}} \quad (21)$$

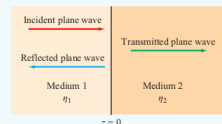
$$\tilde{\mathbf{H}}^r(z) = -(\Gamma E_0^i / \eta_1) e^{jk_1 z} \hat{\mathbf{y}} \quad (22)$$



(a) Normal incidence



(a) Boundary between transmission lines



(b) Boundary between different media

Figure 8-2 Discontinuity between two different transmission lines is analogous to that between two dissimilar media.

Reflection and Transmission at Normal Incidence (Cont.)

Transmitted Wave ($\mathbf{E}^t, \mathbf{H}^t$)

$$\tilde{\mathbf{E}}^t(z) = \tau E_0^i e^{-jk_2 z} \hat{\mathbf{x}} \quad (23)$$

$$\tilde{\mathbf{H}}^t(z) = (\tau E_0^i / \eta_2) e^{-jk_2 z} \hat{\mathbf{y}} \quad (24)$$

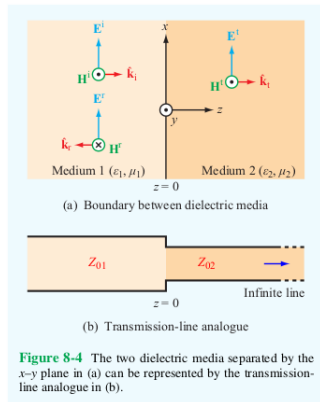
Here, the **reflection coefficient** is

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (25)$$

and the **transmission coefficient** is

$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1} = 1 + \Gamma. \quad (26)$$

When the media are **lossless**, Γ and τ are **real-valued**.



Reflection and Transmission at Normal Incidence (Cont.)

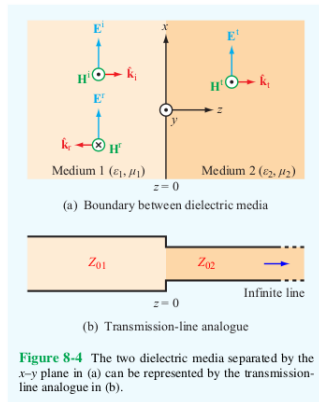
The **electric** and **magnetic fields** in **Medium 1** are

$$\tilde{\mathbf{E}}_1(z) = E_0^i \left(e^{-jk_1 z} + \Gamma e^{jk_1 z} \right) \hat{\mathbf{x}} \quad (27)$$

$$\tilde{\mathbf{H}}_1(z) = \frac{E_0^i}{\eta_1} \left(e^{-jk_1 z} - \Gamma e^{jk_1 z} \right) \hat{\mathbf{y}} \quad (28)$$

and the **average power density** in **Medium 1** is

$$\begin{aligned} \mathbf{S}_{\text{av}1} &= \mathbf{S}_{\text{av}}^i + \mathbf{S}_{\text{av}}^r \\ &= \frac{|E_0^i|^2}{2\eta_1} \left(1 - |\Gamma|^2 \right) \hat{\mathbf{z}}. \end{aligned} \quad (29)$$



Reflection and Transmission at Normal Incidence (Cont.)

The **average power density** in **Medium 2** is

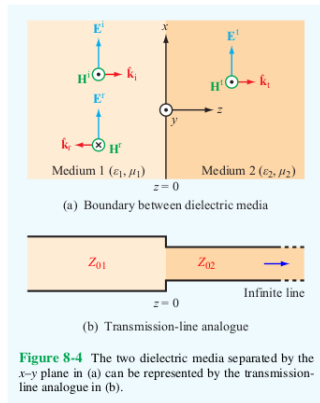
$$\mathbf{S}_{\text{av}2} = \frac{|\tau|^2 |E_0^i|^2}{2\eta_2} \hat{\mathbf{z}}. \quad (30)$$

Since (τ and Γ are real-valued and)

$$\frac{\tau^2}{\eta_2} = \frac{1 - \Gamma^2}{\eta_1}, \quad (31)$$

$$\mathbf{S}_{\text{av}2} = \frac{\tau^2}{\eta_2} \frac{|E_0^i|^2}{2} \hat{\mathbf{z}} = \left(\frac{1 - \Gamma^2}{\eta_1} \right) \frac{|E_0^i|^2}{2} \hat{\mathbf{z}} = \mathbf{S}_{\text{av}1}, \quad (32)$$

as expected from **power conservation**.



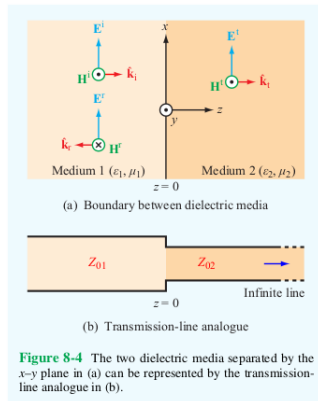
Reflection and Transmission at Normal Incidence (Cont.)

- We explored an example of yellow light normally incident on glass.

Some possible applications of this case are

- windows
- lenses

Note that η_1 , η_2 , Γ , and τ are **independent of frequency** for **lossless media**. However, k_1 and k_2 are **frequency dependent** and $\mathbf{S}_{av1}$ and $\mathbf{S}_{av2}$ only depend on frequency through $|E_0^i|$.



Reflection and Transmission at Normal Incidence (Cont.)

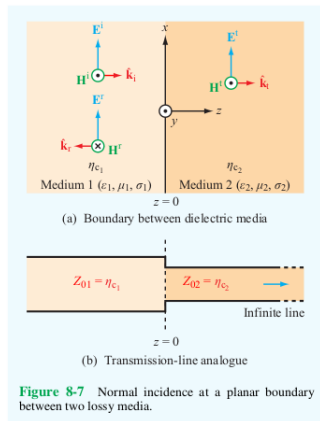
Normal Incidence in Lossy Media

In a **lossy medium**, the **intrinsic impedance** η_c and **propagation constant** $\gamma = \alpha + j\beta$ are **complex-valued**.

Medium 1: $\epsilon_1, \mu_1, \sigma_1$,
 $\epsilon_{c1} = \epsilon_1 - j\sigma_1/\omega$, $\gamma_1 = \alpha_1 + j\beta_1$

$$\tilde{\mathbf{E}}_1(z) = E_0^i (e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}) \hat{\mathbf{x}} \quad (33)$$

$$\tilde{\mathbf{H}}_1(z) = \frac{E_0^i}{\eta_{c1}} (e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}) \hat{\mathbf{y}} \quad (34)$$



Reflection and Transmission at Normal Incidence (Cont.)

Medium 2: $\epsilon_2, \mu_2, \sigma_2$,
 $\epsilon_{c2} = \epsilon_2 - j\sigma_2/\omega, \gamma_2 = \alpha_2 + j\beta_2$

$$\tilde{\mathbf{E}}_2(z) = \tau E_0^i e^{-\gamma_2 z} \hat{\mathbf{x}} \quad (35)$$

$$\tilde{\mathbf{H}}_2(z) = \frac{\tau E_0^i}{\eta_{c2}} e^{-\gamma_2 z} \hat{\mathbf{y}} \quad (36)$$

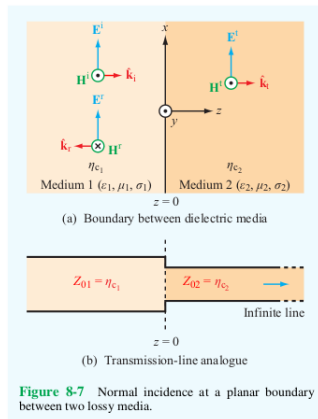
As before, the **reflection and transmission coefficients** are

$$\Gamma = \frac{\eta_{c2} - \eta_{c1}}{\eta_{c2} + \eta_{c1}} \quad (37)$$

and

$$\tau = 1 + \Gamma = \frac{2\eta_{c2}}{\eta_{c2} + \eta_{c1}}. \quad (38)$$

But Γ and τ are **complex-valued**.



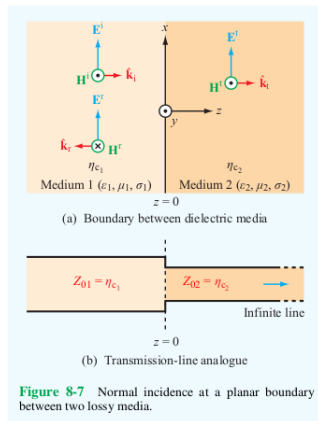
Reflection and Transmission at Normal Incidence (Cont.)

- We considered an example of radio waves normally incident on a metal surface.

Some possible applications of this case are

- mirrors
- shielding
- filters

Note that η_{c1} , η_{c2} , Γ , τ , γ_1 , and γ_2 are **frequency dependent** for **lossy media**.

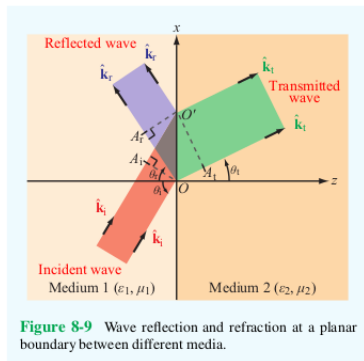


Reflection and Transmission in Geometric Optics

Snell's Laws

For **lossless** media, when the incident wave is traveling in a direction that isn't normal to the boundary, the rays of the waves **travel in the directions below** and **intersect the boundary at the given angles relative to the normal to the boundary**.

- **incident wave:**
direction $\hat{\mathbf{k}}_i$, angle $\theta_i \in [0, \pi/2]$,
- **reflected wave:**
direction $\hat{\mathbf{k}}_r$, angle $\theta_r \in [0, \pi/2]$,
- **transmitted wave:**
direction $\hat{\mathbf{k}}_t$, angle $\theta_t \in [0, \pi/2]$.



Reflection and Transmission in Geometric Optics (Cont.)

- The **index of refraction** of a medium is defined as

$$n = \frac{c}{u_p} = \sqrt{\mu_r \epsilon_r} > 0. \quad (39)$$

- Under these circumstances,

$$\theta_r = \theta_i \quad (\text{Snell's law of reflection}) \quad (40)$$

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \sqrt{\frac{\mu_{r1} \epsilon_{r1}}{\mu_{r2} \epsilon_{r2}}} \quad (\text{Snell's law of refraction}). \quad (41)$$

- For **nonmagnetic materials**, the **law of refraction** is

$$\frac{\sin \theta_t}{\sin \theta_i} = \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \frac{\eta_2}{\eta_1}. \quad (42)$$

- A nonmagnetic material with a **higher index of refraction** is often called **denser** than a nonmagnetic material with a **lower index**.

Reflection and Transmission in Geometric Optics (Cont.)

- When $\theta_i \neq 0$ and $n_2 > n_1$, $\theta_t < \theta_i$, which is called **inward refraction**, and, when $\theta_i \neq 0$ and $n_1 > n_2$, $\theta_t > \theta_i$, which is called **outward refraction**.
- When $n_1 > n_2$, $\theta_t = \pi/2$ when θ_i is at the **critical angle** θ_c , given by

$$\sin \theta_c = n_2/n_1. \quad (43)$$

- If $\mu_1 = \mu_2$,

$$\sin \theta_c = \sqrt{\epsilon_{r2}/\epsilon_{r1}}. \quad (44)$$

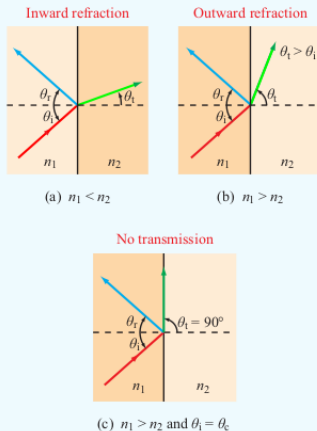
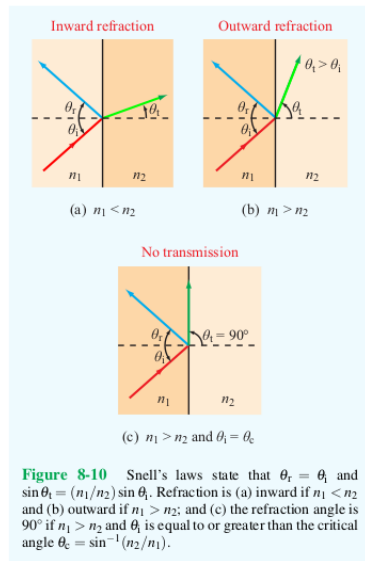


Figure 8-10 Snell's laws state that $\theta_r = \theta_i$ and $\sin \theta_t = (n_1/n_2) \sin \theta_i$. Refraction is (a) inward if $n_1 < n_2$ and (b) outward if $n_1 > n_2$; and (c) the refraction angle is 90° if $n_1 > n_2$ and θ_i is equal to or greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

Reflection and Transmission in Geometric Optics (Cont.)

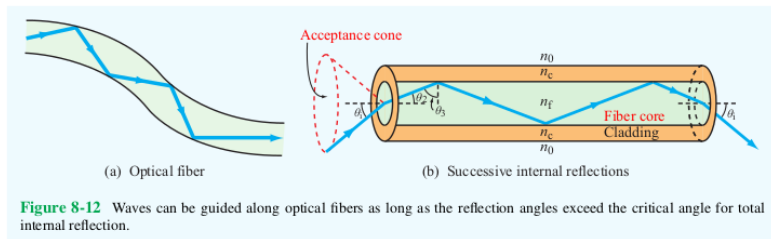
- When $n_1 > n_2$ and $\theta_i \geq \theta_c$, the incident wave is entirely reflected, which is called **total internal reflection**.



Reflection and Transmission in Geometric Optics (Cont.)

Fiber Optics

Total internal reflection can be used to create small-diameter cylinders of dielectric material, typically glass or plastic, that efficiently carry light.



- A fiber consists of an inner **fiber core** with an index of refraction n_f that is higher than the index of refraction n_c of the surrounding **cladding**.

Reflection and Transmission in Geometric Optics (Cont.)

- Light entering the end of the fiber core must have an angle of incidence θ_i satisfying

$$\sin \theta_i \leq \frac{1}{n_0} \left(n_f^2 - n_c^2 \right)^{1/2}, \quad (45)$$

where n_0 is the index of refraction of the medium outside of the end of the fiber, for the ray to be totally internally reflected.

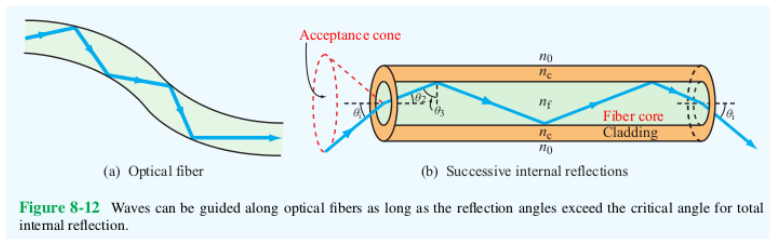
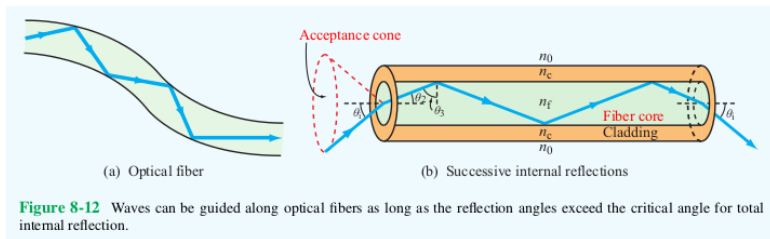


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Reflection and Transmission in Geometric Optics (Cont.)

- The **largest angle** θ_i that produces total internal reflection in the fiber is the **acceptance angle** θ_a , given by

$$\sin \theta_a = \frac{1}{n_0} \left(n_f^2 - n_c^2 \right)^{1/2}. \quad (46)$$



- The cone containing all rays that intersect the fiber core with $\theta_i \leq \theta_a$ is called the **acceptance cone**.

Reflection and Transmission in Geometric Optics (Cont.)

- Rays entering the fiber core at different angles θ_i follow different paths through the fiber. Each path is called a **mode**.
- Different modes travel through the fiber in different amounts of time. This is called **modal dispersion** and distorts pulses sent through the fiber.

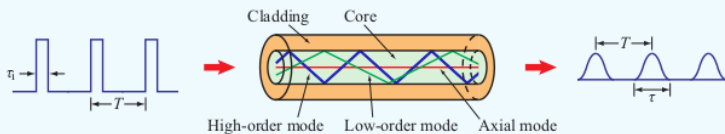


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

- Rectangular pulses of initial width τ_i and period T are spread out into rounded pulses of width τ and pulses are usually created with a period $T \geq 2\tau$ to make sure it's possible to distinguish them.

Reflection and Transmission in Geometric Optics (Cont.)

- For a fiber of length l , the **maximum time delay** will be

$$\tau = \frac{ln_f}{c} \left(\frac{n_f - 1}{n_c} \right) \quad (47)$$

and the **maximum data transmission rate** meeting the requirement $T \geq 2\tau$ will then be

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{cn_c}{2ln_f(n_f - n_c)}. \quad (48)$$

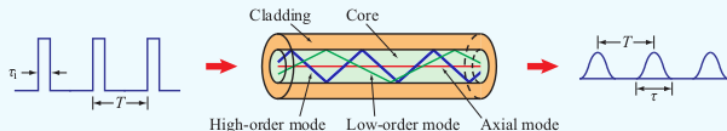
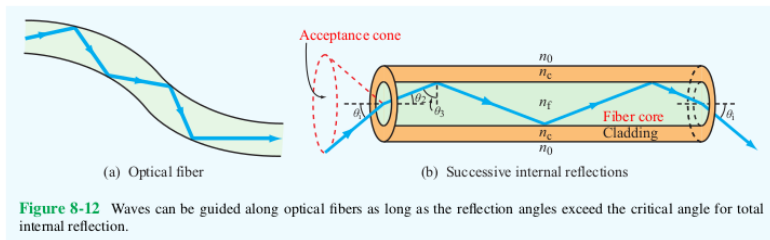


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

Reflection and Transmission in Geometric Optics (Cont.)

- The cone consisting of all of the paths that light at the end of the fiber is allowed to follow is called the **entrance cone**.



Reflection and Transmission in Geometric Optics (Cont.)

- If the entrance cone can be restricted to be narrower than the acceptance cone, with maximum angle $\theta_i(\text{max})$, then the **maximum data transmission rate** can be **increased** to

$$f_p = \frac{c}{2ln_f} \left(\frac{\cos \theta_2(\text{max})}{1 - \cos \theta_2(\text{max})} \right), \quad (49)$$

where

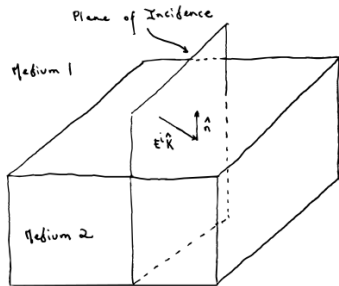
$$\cos \theta_2(\text{max}) = \left(1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_i(\text{max}) \right)^{1/2}. \quad (50)$$

General Model of Reflection and Transmission

Reflection and Transmission at Oblique Incidence

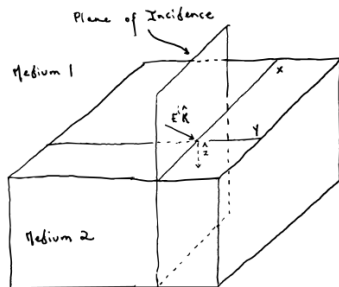
A **more complete picture** of what happens when an **EM wave encounters a boundary between two lossless media** is based on representing the wave as a **sum of two components** – a component ($\mathbf{E}_\perp, \mathbf{H}_\perp$) polarized **perpendicular to the plane of incidence** and a component ($\mathbf{E}_\parallel, \mathbf{H}_\parallel$) polarized **parallel to the plane of incidence**.

The **plane of incidence** is the plane formed by the **normal to the boundary** $\hat{\mathbf{n}}$, the **direction of propagation** $\hat{\mathbf{k}}$ of the EM wave, and the **point where a ray of the wave intersects the boundary**.



General Model of Reflection and Transmission (Cont.)

- The plane of incidence, the boundary between the media, and a point where a ray of the wave intersects the boundary can be used to define a **global Cartesian coordinate system**.



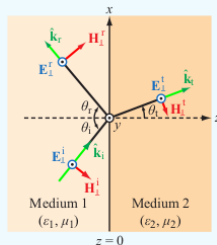
General Model of Reflection and Transmission (Cont.)

- The wave $(\mathbf{E}_{\parallel}, \mathbf{H}_{\parallel})$ is called **transverse magnetic (TM)** since $\mathbf{H}_{\parallel}$ is $\perp$ the plane of incidence and $(\mathbf{E}_{\perp}, \mathbf{H}_{\perp})$ is called **transverse electric (TE)** since $\mathbf{E}_{\perp}$ is $\perp$ the plane of incidence.
- Define $(\mathbf{E}^i, \mathbf{H}^i)$ to be the EM fields of the **incident wave**, $(\mathbf{E}^r, \mathbf{H}^r)$ to be the EM fields of the **reflected wave**, and $(\mathbf{E}^t, \mathbf{H}^t)$ to be the EM fields of the **transmitted wave**. Then

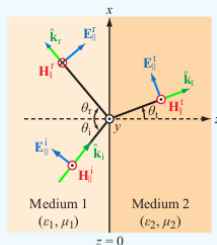
$$\mathbf{E}^i = \mathbf{E}_{\parallel}^i + \mathbf{E}_{\perp}^i, \quad \mathbf{H}^i = \mathbf{H}_{\parallel}^i + \mathbf{H}_{\perp}^i, \quad (51)$$

$$\mathbf{E}^r = \mathbf{E}_{\parallel}^r + \mathbf{E}_{\perp}^r, \quad \mathbf{H}^r = \mathbf{H}_{\parallel}^r + \mathbf{H}_{\perp}^r, \quad (52)$$

$$\mathbf{E}^t = \mathbf{E}_{\parallel}^t + \mathbf{E}_{\perp}^t, \quad \mathbf{H}^t = \mathbf{H}_{\parallel}^t + \mathbf{H}_{\perp}^t. \quad (53)$$



(a) Perpendicular polarization

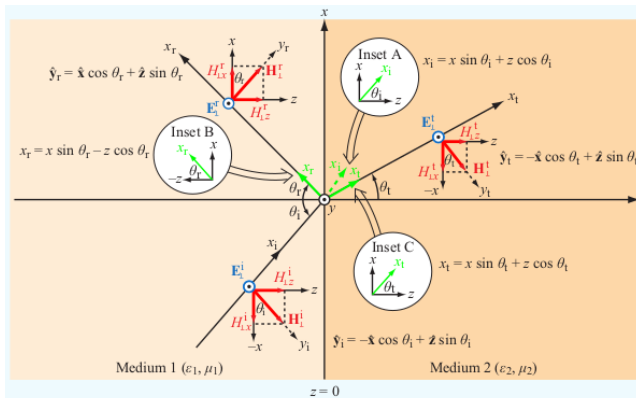


(b) Parallel polarization

General Model of Reflection and Transmission (Cont.)

Perpendicular Polarization

For an incident wave ($\mathbf{E}_\perp^i, \mathbf{H}_\perp^i$) that is **polarized perpendicular** to the plane of incidence,



General Model of Reflection and Transmission (Cont.)

- the phasors of the EM fields of the **incident wave** are

$$\tilde{\mathbf{E}}_{\perp}^i = E_{\perp 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} \hat{\mathbf{y}}, \quad (54)$$

$$\tilde{\mathbf{H}}_{\perp}^i = \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} (-\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i), \quad (55)$$

- the phasors of the EM fields of the **reflected wave** are

$$\tilde{\mathbf{E}}_{\perp}^r = \Gamma_{\perp} E_{\perp 0}^i e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} \hat{\mathbf{y}}, \quad (56)$$

$$\tilde{\mathbf{H}}_{\perp}^r = \Gamma_{\perp} \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} (\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i), \quad (57)$$

- and phasors of the the EM fields of the **transmitted wave** are

$$\tilde{\mathbf{E}}_{\perp}^t = \tau_{\perp} E_{\perp 0}^i e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} \hat{\mathbf{y}}, \quad (58)$$

$$\tilde{\mathbf{H}}_{\perp}^t = \tau_{\perp} \frac{E_{\perp 0}^i}{\eta_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} (-\hat{\mathbf{x}} \cos \theta_t + \hat{\mathbf{z}} \sin \theta_t), \quad (59)$$

General Model of Reflection and Transmission (Cont.)

where the **Fresnel reflection coefficient for perpendicular polarization**

$$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}, \quad (60)$$

the **Fresnel transmission coefficient for perpendicular polarization**

$$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}, \quad (61)$$

and **Snell's law of refraction**

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{k_1}{k_2} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \frac{n_1}{n_2} \quad (62)$$

determines θ_t from θ_i and the constitutive parameters of the two media.

General Model of Reflection and Transmission (Cont.)

- The relationships between the amplitudes of the reflected and transmitted waves, $E_{\perp 0}^r$ and $E_{\perp 0}^t$, and the angles of reflection and transmission, θ_r , and θ_t , and the 2 parameters $E_{\perp 0}^i$ and θ_i were determined using the **tangential boundary conditions**.
- These imply the **phase-matching condition**

$$k_1 \sin \theta_i = k_1 \sin \theta_r = k_2 \sin \theta_t. \quad (63)$$

- Aside from Snell's law of refraction, the phase-matching condition implies **Snell's law of reflection**

$$\theta_r = \theta_i. \quad (64)$$

- Reflecting the **balance of power across the boundary**

$$\tau_{\perp} = 1 + \Gamma_{\perp}. \quad (65)$$

General Model of Reflection and Transmission (Cont.)

Special cases:

- When **Medium 2 is a perfect conductor**, $\eta_2 = 0$, $\Gamma_{\perp} = -1$, and $\tau_{\perp} = 0$. So, the wave is perfectly reflected.
- When the media are **nonmagnetic**, $\mu_1 = \mu_2 = \mu_0$ and

$$\Gamma_{\perp} = \frac{\cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}. \quad (66)$$

Here, $\epsilon_2/\epsilon_1 = (n_2/n_1)^2$. So, $\Gamma_{\perp}$ can be written in terms of the indices of refraction.

General Model of Reflection and Transmission (Cont.)

Parallel Polarization

For an incident wave $(\mathbf{E}_{\parallel}^i, \mathbf{H}_{\parallel}^i)$ that is **polarized parallel** to the plane of incidence, the phasors of the EM fields of the **incident wave** are

$$\tilde{\mathbf{E}}_{\parallel}^i = E_{\parallel 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} \cdot (\hat{\mathbf{x}} \cos \theta_i - \hat{\mathbf{z}} \sin \theta_i), \quad (67)$$

$$\tilde{\mathbf{H}}_{\parallel}^i = \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} \hat{\mathbf{y}}. \quad (68)$$

The **tangential boundary conditions** imply Snell's laws and relationships between the amplitudes and angles of the reflected and transmitted waves and those of the incident wave.

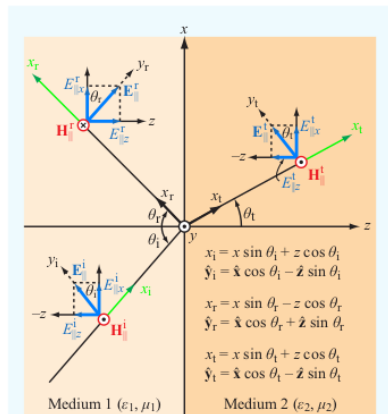


Figure 8-16 Parallel-polarized plane wave incident at an angle θ_i upon a planar boundary.

General Model of Reflection and Transmission (Cont.)

The phasors of the EM fields of the **reflected wave** are

$$\tilde{\mathbf{E}}_{\parallel}^r = \Gamma_{\parallel} E_{\parallel 0}^i e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} (\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i), \quad (69)$$

$$\tilde{\mathbf{H}}_{\parallel}^r = -\Gamma_{\parallel} \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} \hat{\mathbf{y}}, \quad (70)$$

and the phasors of the EM fields of the **transmitted wave** are

$$\tilde{\mathbf{E}}_{\parallel}^t = \tau_{\parallel} E_{\parallel 0}^i e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} (\hat{\mathbf{x}} \cos \theta_t - \hat{\mathbf{z}} \sin \theta_t), \quad (71)$$

$$\tilde{\mathbf{H}}_{\parallel}^t = \tau_{\parallel} \frac{E_{\parallel 0}^i}{\eta_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} \hat{\mathbf{y}}, \quad (72)$$

where the **Fresnel reflection coefficient for parallel polarization** is

$$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad (73)$$

General Model of Reflection and Transmission (Cont.)

the **Fresnel transmission coefficient for parallel polarization** is

$$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad (74)$$

and **Snell's law of refraction** implies

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{k_1}{k_2} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \frac{n_1}{n_2}. \quad (75)$$

Note that the boundary conditions also imply **Snell's law of reflection**

$$\theta_r = \theta_i. \quad (76)$$

In this case,

$$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}. \quad (77)$$

General Model of Reflection and Transmission (Cont.)

Special cases:

- When **Medium 2 is a perfect conductor**, $\eta_2 = 0$ and $\Gamma_{\parallel} = -1$ and $\tau_{\parallel} = 0$.
- When the media are **nonmagnetic**, $\mu_1 = \mu_2 = \mu_0$ and

$$\Gamma_{\parallel} = \frac{-(\epsilon_2/\epsilon_1) \cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}. \quad (78)$$

General Model of Reflection and Transmission (Cont.)

Brewster Angle

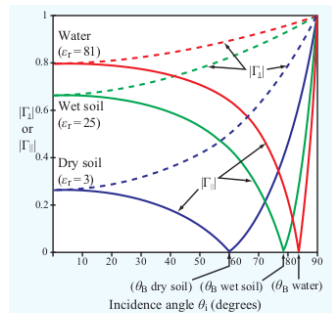
There is an angle of incidence $\theta_{B\parallel}$ for EM waves **polarized parallel to the plane of incidence** where $\Gamma_{\parallel} = 0$. This angle is called the **Brewster angle**. It satisfies

$$\sin \theta_{B\parallel} = \sqrt{\frac{1 - (\epsilon_1 \mu_2 / \epsilon_2 \mu_1)}{1 - (\epsilon_1 / \epsilon_2)^2}}. \quad (79)$$

When $\mu_1 = \mu_2$,

$$\theta_{B\parallel} = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}. \quad (80)$$

When $\theta_i = \theta_{B\parallel}$, the parallel polarized component is **completely transmitted** into Medium 2 and **the reflected wave is perpendicularly polarized**, thereby “polarizing” the wave. So, the Brewster angle is also called the **polarizing angle**.



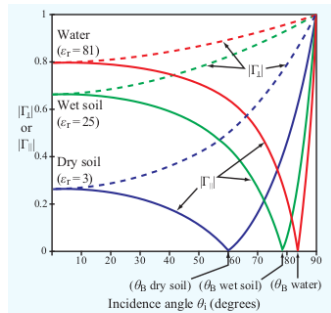
General Model of Reflection and Transmission (Cont.)

When $\mu_1 \neq \mu_2$, there is also a Brewster angle $\theta_{B\perp}$ for EM waves **polarized perpendicular to the plane of incidence** where $\Gamma_{\perp} = 0$. It is given by

$$\sin \theta_{B\perp} = \sqrt{\frac{1 - (\mu_1 \epsilon_2 / \mu_2 \epsilon_1)}{1 - (\mu_1 / \mu_2)^2}}. \quad (81)$$

Other Notable Angles of Incidence:

- At **normal incidence** ($\theta_i = 0$), $\Gamma_{\perp} = \Gamma_{\parallel}$.
- When $\theta_i = 90^\circ$, which is called **grazing incidence**, $|\Gamma_{\perp}| = |\Gamma_{\parallel}| = 1$.



General Model of Reflection and Transmission (Cont.)

Suppose a circular beam of EM radiation shines on the boundary between two lossless media.

The **reflectivity** (or **reflectance**) R is the ratio of the reflected power over the incident power. For the **perpendicularly polarized** component, the reflectivity is

$$R_{\perp} = \frac{P_{\perp}^r}{P_{\perp}^i} = |\Gamma_{\perp}|^2. \quad (82)$$

In the case of **parallel polarization**, the reflectivity is

$$R_{\parallel} = \frac{P_{\parallel}^r}{P_{\parallel}^i} = |\Gamma_{\parallel}|^2. \quad (83)$$

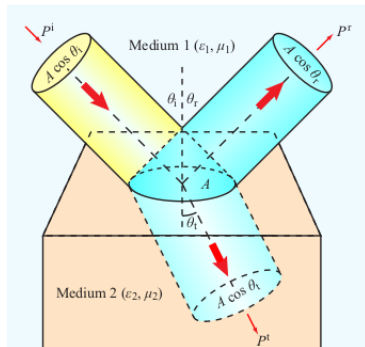


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

General Model of Reflection and Transmission (Cont.)

The **transmissivity** (or **transmittance**) T is the ratio of the transmitted power over the incident power. For the **perpendicularly polarized** component, the transmissivity is

$$T_{\perp} = \frac{P_{\perp}^t}{P_{\perp}^i} = |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right). \quad (84)$$

In the case of **parallel polarization**, the transmissivity is

$$T_{\parallel} = \frac{P_{\parallel}^t}{P_{\parallel}^i} = |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right). \quad (85)$$

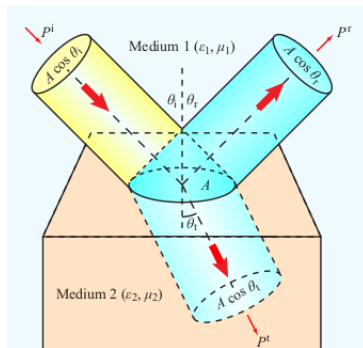


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

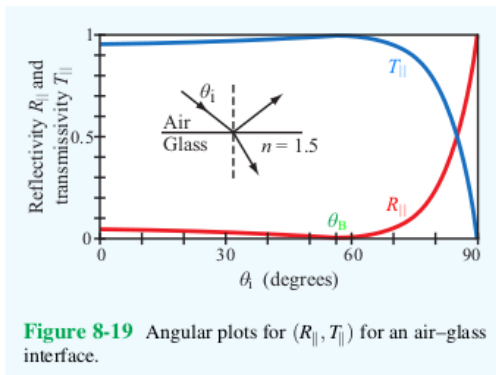
General Model of Reflection and Transmission (Cont.)

Conservation of power implies

$$R_{\perp} + T_{\perp} = 1, \quad R_{\parallel} + T_{\parallel} = 1, \quad (86)$$

which are the same as

$$|\Gamma_{\perp}|^2 + |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = 1, \quad |\Gamma_{\parallel}|^2 + |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = 1. \quad (87)$$



General Model of Reflection and Transmission (Cont.)

Table 8-2 Expressions for Γ , τ , R , and T for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2 . Angles θ_i and θ_t are the angles of incidence and transmission, respectively.

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Transmission coefficient	$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
Reflectivity	$R = \Gamma ^2$	$R_{\perp} = \Gamma_{\perp} ^2$	$R_{\parallel} = \Gamma_{\parallel} ^2$
Transmissivity	$T = \tau ^2 \left(\frac{\eta_1}{\eta_2} \right)$	$T_{\perp} = \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$T_{\parallel} = \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
Relation of R to T	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$
Notes: (1) $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; (2) $\eta_1 = \sqrt{\mu_1 / \epsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2 / \epsilon_2}$; (4) for nonmagnetic media, $\eta_2 / \eta_1 = n_1 / n_2$.			

Waveguides and Cavities

Waveguides

A **waveguide** is a **higher-order transmission lines** that carries EM waves where one of $\mathbf{E}$ or $\mathbf{H}$ is orthogonal to the direction of propagation $\hat{\mathbf{k}}$ but not both.

An **optical fiber** is another example of this type of transmission line. **Metal waveguides use reflection at the boundary between a dielectric medium and a conductor** to direct EM waves.

A **mode** is

- **transverse electric (TE)** if $\mathbf{E}$ is orthogonal to $\hat{\mathbf{k}}$ but not $\mathbf{H}$,
- **transverse magnetic (TM)** if $\mathbf{H}$ is orthogonal to $\hat{\mathbf{k}}$ but not $\mathbf{E}$.

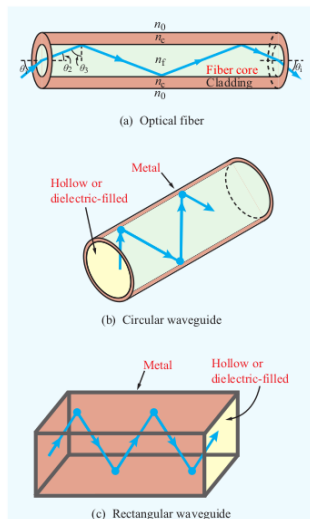


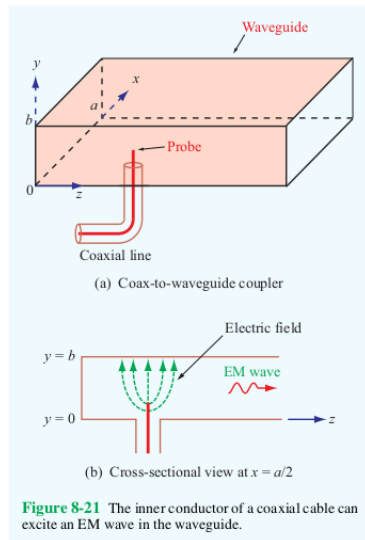
Figure 8-20 Wave travel by successive reflections in (a) an optical fiber, (b) a circular metal waveguide, and (c) a rectangular metal waveguide.

Waveguides and Cavities (Continued)

Signals can be **transferred between coaxial cables and metal waveguides** as shown in the figure. **Coaxial cables** are appropriate for transmitting frequencies **below 30 GHz** but **metal waveguides** are a good alternative for **high-power RF frequencies from 5 to 100 GHz**.

Rectangular Metal Waveguides

We explored the **modes**, i.e. types of EM waves, that will propagate through a **rectangular metal waveguide** that is filled with a **lossless dielectric material**, which is often air. The waveguide has **width a** and **height b** , where $a > b$.

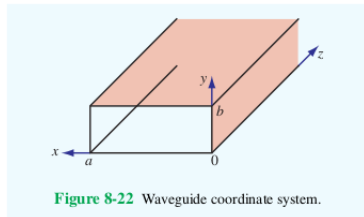


Waveguides and Cavities (Continued)

The **Cartesian coordinate system** shown in the adjacent figure is appropriate for this purpose, where the **direction of propagation** of the EM waves is $\hat{z}$. In this coordinate system, the phasors of the electric and magnetic fields have the form

$$\tilde{\mathbf{E}} = \tilde{E}_x \hat{x} + \tilde{E}_y \hat{y} + \tilde{E}_z \hat{z} \quad (88)$$

$$\tilde{\mathbf{H}} = \tilde{H}_x \hat{x} + \tilde{H}_y \hat{y} + \tilde{H}_z \hat{z}. \quad (89)$$



The modes can be divided into

- **transverse magnetic modes** with $\tilde{\mathbf{H}}$ orthogonal to $\hat{z}$, i.e. $\tilde{H}_z = 0$,
- **transverse electric modes** with $\tilde{\mathbf{E}}$ orthogonal to $\hat{z}$, i.e. $\tilde{E}_z = 0$.

Only EM waves in these modes propagate through the waveguide.

Waveguides and Cavities (Continued)

Assuming that the waves propagate in the $+z$ direction, using Maxwell's equations, and imposing appropriate boundary conditions, the **m nth transverse magnetic mode TM_{mn}** has components

$$\tilde{E}_x(x, y, z) = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (90)$$

$$\tilde{E}_y(x, y, z) = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (91)$$

$$\tilde{E}_z(x, y, z) = E_0 \sin \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (92)$$

$$\tilde{H}_x(x, y, z) = \frac{j\omega\epsilon}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (93)$$

$$\tilde{H}_y(x, y, z) = -\frac{j\omega\epsilon}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (94)$$

$$\tilde{H}_z(x, y, z) = 0, \quad (95)$$

where $m = 1, 2, \dots$ and $n = 1, 2, \dots$,

Waveguides and Cavities (Continued)

and the mn^{th} **transverse electric mode** $\mathbf{TE}_{mn}$ has components

$$\tilde{E}_x(x, y, z) = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \quad (96)$$

$$\tilde{E}_y(x, y, z) = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \quad (97)$$

$$\tilde{E}_z(x, y, z) = 0 \quad (98)$$

$$\tilde{H}_x(x, y, z) = \frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \quad (99)$$

$$\tilde{H}_y(x, y, z) = \frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \quad (100)$$

$$\tilde{H}_z(x, y, z) = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (101)$$

where $m = 1, 2, \dots$ and $n = 0, 1, \dots$, assuming that $a > b$.

Waveguides and Cavities (Continued)

Here, E_0 and H_0 are **amplitudes**, the **cutoff wavenumber** is

$$k_c^2 = k^2 - \beta^2 = \omega^2 \mu \epsilon - \beta^2, \quad (102)$$

the **unbounded-medium wavenumber** is

$$k = \omega \sqrt{\mu \epsilon}, \quad (103)$$

and the **phase constant** is

$$\beta = \sqrt{\omega^2 \mu \epsilon - (m\pi/a)^2 - (n\pi/b)^2}. \quad (104)$$

For each mode, with specific values for m and n , the lowest frequency for a wave that will propagate through the waveguide is the **cutoff frequency**

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \quad (105)$$

where

$$u_{p0} = 1/\sqrt{\mu \epsilon} \quad (106)$$

is the **speed of propagation in an unbounded medium**.

Waveguides and Cavities (Continued)

The phase constant can be expressed in terms of the cutoff frequency

$$\beta = \frac{\omega}{u_{p0}} \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2} \quad (107)$$

and the **phase velocity** or **speed of propagation** of a TM or TE wave in the waveguide is

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad (108)$$

These both **depend on the frequency** f of the EM wave.

The **wave impedance** in the waveguide is

$$Z_{\text{TM}} = \eta \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2} \quad (\text{TM modes}) \quad (109)$$

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}} \quad (\text{TE modes}). \quad (110)$$

Waveguides and Cavities (Continued)

The **dominant mode** is the mode with the **lowest cutoff frequency**. This is TM_{11} for the TM modes and TE_{10} for the TE modes. The adjacent figure shows the fields of the mode TM_{11} .

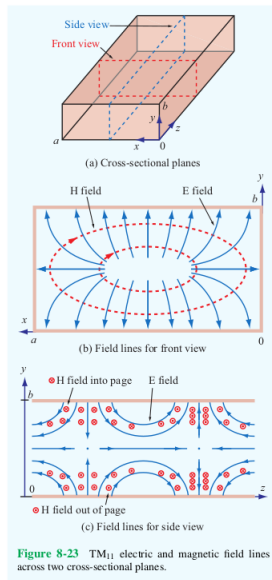


Figure 8-23 TM_{11} electric and magnetic field lines across two cross-sectional planes.

Waveguides and Cavities (Continued)

Table 8-3 Wave properties for TE and TM modes in a rectangular waveguide with dimensions $a \times b$, filled with a dielectric material with constitutive parameters ϵ and μ . The TEM case, shown for reference, pertains to plane-wave propagation in an unbounded medium.

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/Z_{TE}$ $\tilde{H}_y = \tilde{E}_x/Z_{TE}$ $\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	$\tilde{E}_x = \frac{-j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{H}_x = -\tilde{E}_y/Z_{TM}$ $\tilde{H}_y = \tilde{E}_x/Z_{TM}$ $\tilde{H}_z = 0$ $Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	$\tilde{E}_x = E_{x0} e^{-j\beta z}$ $\tilde{E}_y = E_{y0} e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/\eta$ $\tilde{H}_y = \tilde{E}_x/\eta$ $\tilde{H}_z = 0$ $\eta = \sqrt{\mu/\epsilon}$
<i>Properties Common to TE and TM Modes</i> $f_c = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$ $\beta = k \sqrt{1 - (f_c/f)^2}$ $u_p = \frac{\omega}{\beta} = u_{p0} / \sqrt{1 - (f_c/f)^2}$		$f_c = \text{not applicable}$ $k = \omega \sqrt{\mu\epsilon}$ $u_{p0} = 1/\sqrt{\mu\epsilon}$

Waveguides and Cavities (Continued)

Propagation Velocities

Information is **encoded** into EM waves by modifying their **amplitude**, **frequency**, or **phase**. A modified wave has an **envelope**, which is the shape encapsulating the wave, and consists of the **frequency components** associated with its Fourier representation. The envelope propagates at the **group velocity**

$$u_g = 1/(d\beta/d\omega). \quad (111)$$

For **TM** or **TE** mode mn in a **rectangular waveguide**,

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \quad (112)$$

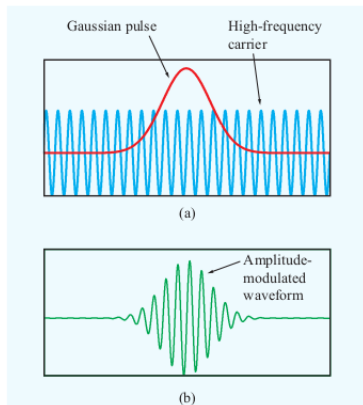


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

Waveguides and Cavities (Continued)

and the individual frequency components propagate at the **phase velocity**

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad (113)$$

Here,

$$u_p u_g = u_{p0}^2 \quad (114)$$

and, for the **waves that propagate through the waveguide**, $f > f_{mn}$. So,

- $u_p > u_{p0}$ and $u_g < u_{p0}$
- $\lim_{f \rightarrow \infty} u_p = \lim_{f \rightarrow \infty} u_g = u_{p0}$.

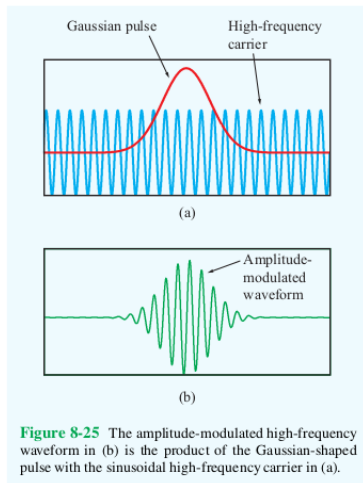


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

Waveguides and Cavities (Continued)

An ω - β **diagram** shows **how TM or TE modes propagate** in a waveguide compared to **how they would propagate in an unbounded medium**. At a point on a curve in this diagram,

- $u_p = \omega/\beta$,
- $u_g = d\omega/d\beta$ is the slope of the curve.

A curve for mode mn has slope 0 and $\omega = 2\pi f_{mn}$ at $\beta = 0$. So, at this point, $u_g = 0$ and $u_p \rightarrow \infty$ as $\beta \rightarrow 0^+$.

The **TEM line** shows ω vs β when the medium is **unbounded**. On this line, $u_g = u_p = u_{p0}$ and the curves for all modes approach the TEM line as $f \rightarrow \infty$.

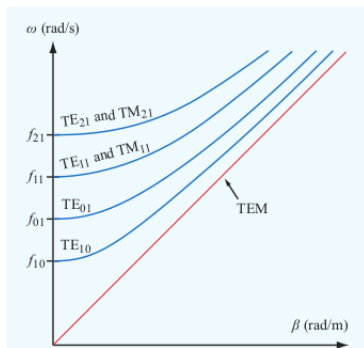


Figure 8-26 ω - β diagram for TE and TM modes in a hollow rectangular waveguide. The straight line pertains to propagation in an unbounded medium or on a TEM transmission line.

Waveguides and Cavities (Continued)

The above results pertain to **straight-line travel through the medium** in a waveguide and don't consider the consequences of reflection. Determining the **overall propagation velocity** requires accounting for the **zigzagging** of the path that a wave follows through the waveguide.

Waveguides and Cavities (Continued)

Cavity Resonators

A **cavity** is a waveguide where the open ends have been replaced with metal walls. For the $a \times b \times d$ rectangular cavity shown in the figure with given the coordinate system, the boundary conditions imply

$$\beta = \frac{p\pi}{d}, \quad p = 1, 2, \dots \quad (115)$$

A mode in the cavity is specified by the integers (m, n, p) . The cavity only sustains a mode at the **resonant frequency**

$$f_{mnp} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}. \quad (116)$$

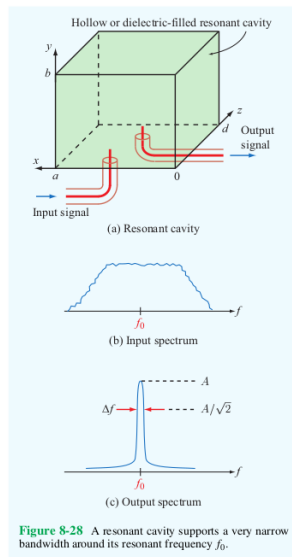


Figure 8-28 A resonant cavity supports a very narrow bandwidth around its resonant frequency f_0 .

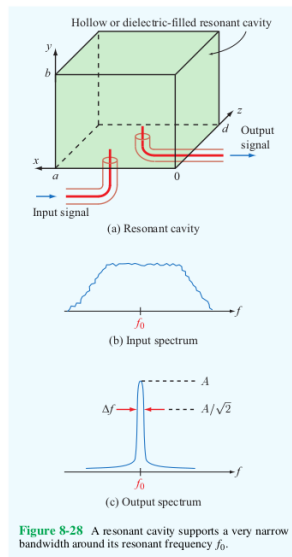
Waveguides and Cavities (Continued)

For the TM modes, m and n start at 1 and p starts at 0. For the TE modes, m and n start at 0 and p starts at 1.

Although, in the ideal case, only the wave with frequency f_{mnp} persists in the cavity, a physical cavity will sustain waves with a small range of frequencies centered at f_{mnp} . A measure of this is the **quality factor**

$$Q \approx \frac{f_{mnp}}{\Delta f}, \quad (117)$$

where Δf is measured at the point on the peak that is $f_{mnp}/\sqrt{2}$. Here, $\frac{\Delta f}{f_{mnp}}$ is the **normalized bandwidth** of the cavity.



Chapter 9: Radiation and Antennas

An antenna

- **transmits** – converts a guided EM wave into a wave that propagates in an unbounded medium,
- or **receives** – converts an EM wave propagating in an unbounded medium into a guided EM wave.

The **size** of an antenna is measured in units of the **wavelength** λ it is intended to transmit or receive.

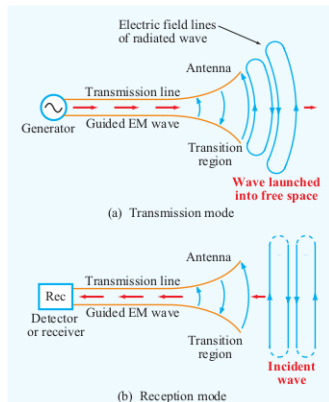
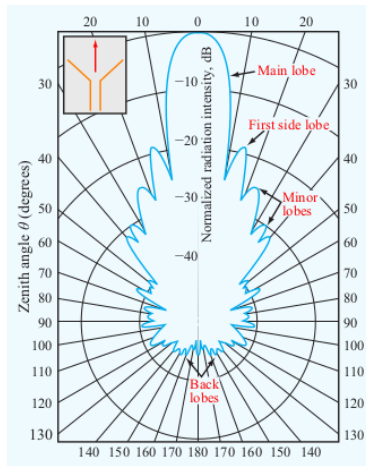


Figure 9-1 Antenna as a transducer between a guided electromagnetic wave and a free-space wave, for both transmission and reception.

Radiation and Antennas (Continued)

Some ways to **characterize** an antenna:

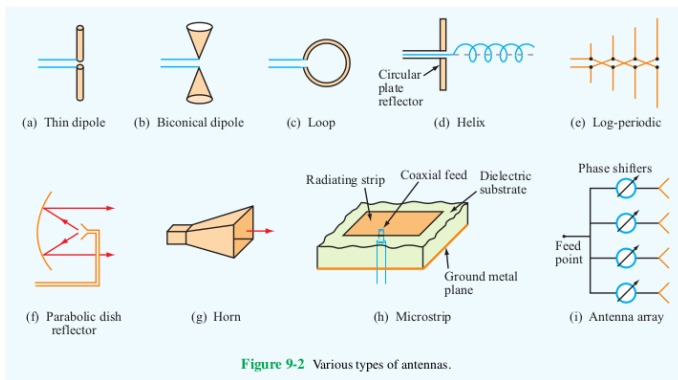
- **antenna radiation pattern** – The **spatial distribution of relative power** that the antenna transmits. Also called the **radiation pattern** or **antenna pattern**.
- **antenna polarization** – The polarization of the emitted wave when the antenna transmits.
- **antenna impedance** associated with the transfer of power from the wave generator to the antenna.



The **relative sensitivity as a receiver** to incoming EM waves from all directions usually **matches its antenna pattern**, which is called **reciprocity**. All antennas considered are reciprocal.

Radiation and Antennas (Continued)

Isotropic Antenna – Hypothetical antenna that radiates (or receives) **uniformly in all directions**, used as a reference.



Antennas are characterized by their radiation source:

- **current sources** – time-varying currents produce the EM waves
- **aperture fields** – EM fields generated across an opening/aperture

Radiation and Antennas (Continued)

Antenna Array – Acts like a single antenna, where the **antenna pattern can be adjusted** to produce a **specific shape** by controlling the waves emitted by each antenna.

Far field – Region **far enough away** from an antenna such that the **wavefronts are planar**. Only investigated the far field.

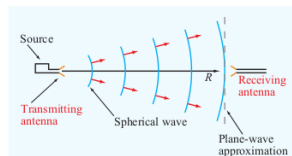


Figure 9-3 Far-field plane-wave approximation.

Hertzian Dipole

Thin, linear conductor with length $l \ll \lambda$ ($l \leq \lambda/50$), where λ is the wavelength of the EM waves it is intended to produce.

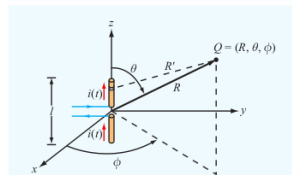


Figure 9-4 Short dipole placed at the origin of a spherical coordinate system.

Radiation and Antennas (Continued)

The current in a Hertzian dipole is

$$i(t) = I_0 \cos \omega t \text{ (A)} = \text{Re} \left\{ I_0 e^{j\omega t} \right\}. \quad (118)$$

So,

$$\tilde{I} = I_0. \quad (119)$$

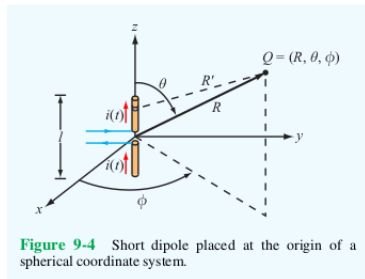
In the **far field**, where $R \gg \lambda$,

$$\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0 \quad (120)$$

$$\tilde{E}_\theta = \frac{jI_0 \ell k \eta_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta = \frac{jk\eta_0}{4\pi} \tilde{I}(z) \ell \left(\frac{e^{-jkR}}{R} \right) \sin \theta \text{ (V/m)} \quad (121)$$

$$\tilde{H}_\phi = \tilde{E}_\theta / \eta_0 \text{ (A/m)} \quad (122)$$

and the EM wave is a TEM plane wave propagating in the $\hat{\mathbf{R}}$ direction.



Radiation and Antennas (Continued)

The **average power density** that a Hertzian dipole emits is

$$S(R, \theta) = S_0 \sin^2 \theta \quad (\text{W/m}^2), \quad (123)$$

where

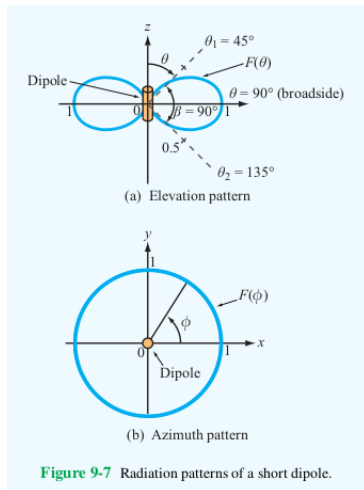
$$S_{\max} = S_0 = \frac{\eta_0 k^2 I_0^2 \ell^2}{32\pi^2 R^2} \quad (124)$$

is the maximum power density at a given distance R .

The **normalized radiation intensity**

$$F(\theta, \phi) = \frac{S(R, \theta, \phi)}{S_{\max}} = F(\theta) = \sin^2 \theta \quad (125)$$

specifies the antenna pattern.



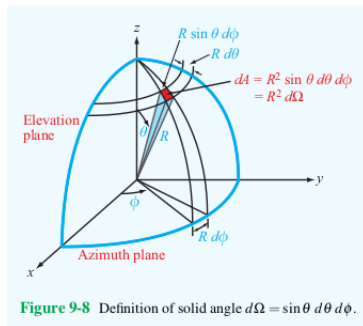
Radiation and Antennas (Continued)

Antenna Radiation Characteristics

An **antenna pattern** represents the **far-field directional properties** of an antenna at a given distance. It's the average power that the antenna radiates through a sphere centered on the antenna in the far-field.

- **Total Radiated Power**

$$P_{\text{rad}} = R^2 S_{\text{max}} \iint_{4\pi} F(\theta, \phi) d\Omega \quad (126)$$



Radiation and Antennas (Continued)

- **Antenna Pattern** – normalized radiation intensity $F(\theta, \phi)$
 - Can be plotted in 3D.
 - Can be plotted as 2D cross-sections.
 - elevation plane:** ϕ is constant
 - azimuth plane:** θ is constant

Usually plotted on a **decibel scale**
 $F \text{ (dB)} = 10 \log F$.

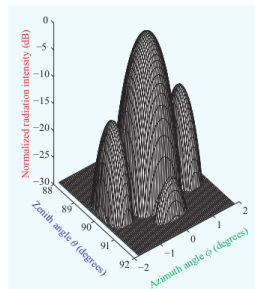
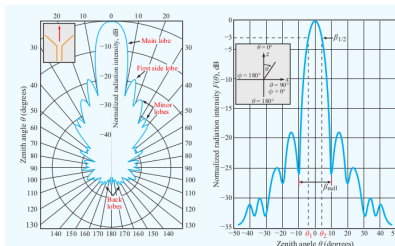


Figure 9-9 Three-dimensional pattern of a narrow-beam antenna.



Radiation and Antennas (Continued)

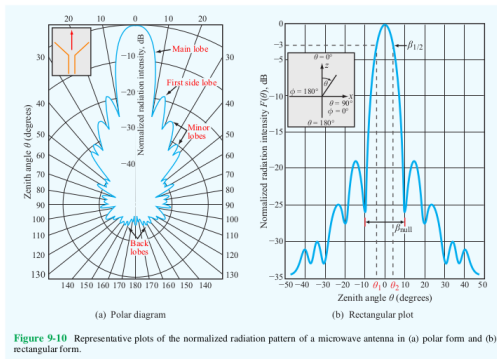


Figure 9-10 Representative plots of the normalized radiation pattern of a microwave antenna in (a) polar form and (b) rectangular form.

- **main lobe** – peak in the antenna pattern containing the largest amount of energy
- **side lobes** – other less prominent peaks in the antenna pattern
- **back lobes** – peaks that point away from the main lobe

Radiation and Antennas (Continued)

• Beam Dimensions

• Pattern Solid Angle

$$\Omega_p = \int \int_{4\pi} F(\theta, \phi) d\Omega \quad (\text{sr}) \quad (127)$$

Smaller Ω_p = **more directional**.

Largest Ω_p is 4π , which is the value for an isotropic antenna.

• Beamwidth

half-power or 3 dB beamwidth β

– angular width between points where the power in the main lobe is half its maximum.

null beamwidth β_{null} – angular width between first nulls on opposite sides of the peak

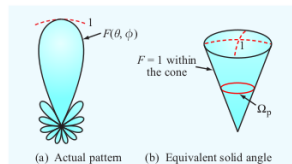
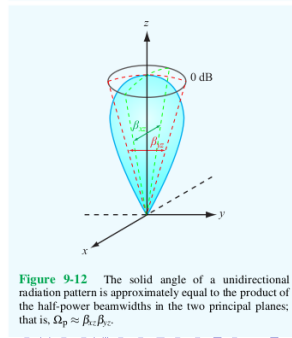


Figure 9-11 The pattern solid angle Ω_p defines an equivalent cone over which all the radiation of the actual antenna is concentrated with uniform intensity equal to the maximum of the actual pattern.



Radiation and Antennas (Continued)

- **Antenna Directivity**

$$D = \frac{F_{\max}}{F_{\text{av}}} = \frac{S_{\max}}{S_{\text{av}}} = \frac{4\pi}{\Omega_p} \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \quad (128)$$

Larger D means more directional. Min value = 1 for isotropic antenna.

- **Antenna Gain**

The **radiation efficiency** of an antenna is

$$\xi = \frac{P_{\text{rad}}}{P_t}, \quad (129)$$

where P_t is the power that the generator applies to the antenna.

Then the antenna **gain**, which accounts for lost power, is

$$G = \frac{4\pi R^2 S_{\max}}{P_t} = \xi D. \quad (130)$$

For a lossless antenna, $\xi = 1$ and $G = D$.

Radiation and Antennas (Continued)

- **Radiation Resistance**

$$R_{\text{rad}} = \frac{2}{I_0^2} P_{\text{rad}} \quad (131)$$

Also, the **radiation efficiency** can be written as

$$\xi = \frac{P_{\text{rad}}}{P_t} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}, \quad (132)$$

where

$$P_t = P_{\text{rad}} + P_{\text{loss}}, \quad R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}, \quad \text{and} \quad R_{\text{loss}} = \frac{2}{I_0^2} P_{\text{loss}}. \quad (133)$$

Radiation and Antennas (Continued)

Half-Wave Dipole Antenna

- Length:

$$\ell = \lambda/2 \quad (134)$$

- Current:

$$i(t) = I_0 \cos(kz) \cos(\omega t) \quad (135)$$

$$\tilde{I}(z) = I_0 \cos(kz) \quad (136)$$

where

$$-\lambda/4 \leq z \leq \lambda/4 \quad (137)$$

and the **wavenumber**

$$k = 2\pi/\lambda. \quad (138)$$

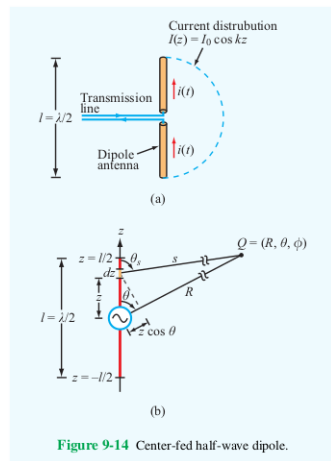


Figure 9-14 Center-fed half-wave dipole.

Radiation and Antennas (Continued)

- **Phasors of the Electric and Magnetic Fields:**

In the far field,

$$\tilde{E}_\theta = j60I_0 \left(\frac{e^{-jkR}}{R} \right) \left(\frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right) \quad (139)$$

$$\tilde{H}_\phi = \tilde{E}_\theta / \eta_0 \quad (140)$$

with $\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0$.

- **Average Power Density:**

$$S(R, \theta) = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos^2\left(\frac{\pi}{2} \cos\theta\right)}{\sin^2\theta} \right) \quad (141)$$

The maximum average power density is

$$S_{\max} = S_0 = \frac{15I_0^2}{\pi R^2}. \quad (142)$$

Radiation and Antennas (Continued)

- **Normalized Radiation Intensity:**

$$F(\theta, \phi) = F(\theta) = \frac{S(R, \theta)}{S_0} = \left(\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right)^2 \quad (143)$$

- **Total Radiated Power:**

$$P_{\text{rad}} = 36.6 I_0^2 \text{ (W)} \quad (144)$$

- **Directivity:**

$$D = 1.64, \quad D \text{ (dB)} = 2.15 \text{ dB} \quad (145)$$

- **Radiation Resistance:**

$$R_{\text{rad}} = 73.2 \text{ } (\Omega) \quad (146)$$

- **Input Impedance:**

$$Z_{\text{in}} \approx R_{\text{rad}} + jX_{\text{in}}, \quad (147)$$

where $X_{\text{in}} = 0$ when the antenna is shortened slightly.

Radiation and Antennas (Continued)

Quarter-Wave Monopole Antenna

This antenna has **length**

$$\ell = \lambda/4 \quad (148)$$

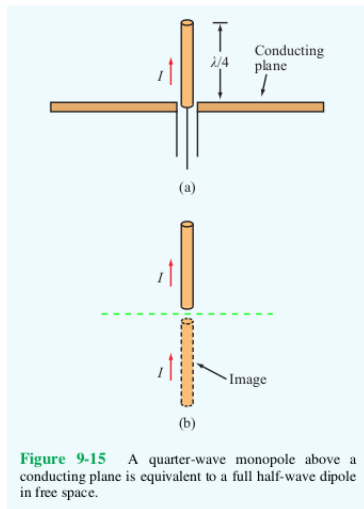
and is positioned **above a conducting ground plane**.

- **EM Field Phasors:**

$$\begin{aligned} \tilde{E}_\theta &= j60I_0 \left(\frac{e^{-jkR}}{R} \right) \left(\frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right), \\ \tilde{H}_\phi &= \tilde{E}_\theta / \eta_0, \end{aligned} \quad (149)$$

and $\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0$ for

$$0 \leq \theta \leq \pi/2. \quad (150)$$



Radiation and Antennas (Continued)

- **Normalized Radiation Intensity:**

$$F(\theta) = \left(\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right)^2, \quad (151)$$

- **Total Radiated Power:**

$$P_{\text{rad}} = \frac{1}{2}(36.6 I_0^2) = 18.3 I_0^2 \quad (\text{W}), \quad (152)$$

- **Radiation Resistance:**

$$R_{\text{rad}} = \frac{1}{2}(73.2) = 36.6 \quad (\Omega). \quad (153)$$

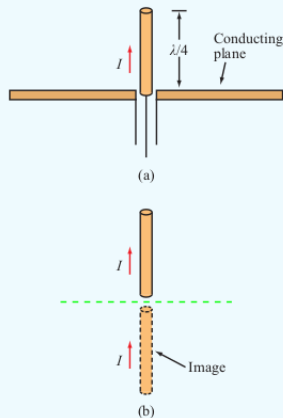


Figure 9-15 A quarter-wave monopole above a conducting plane is equivalent to a full half-wave dipole in free space.

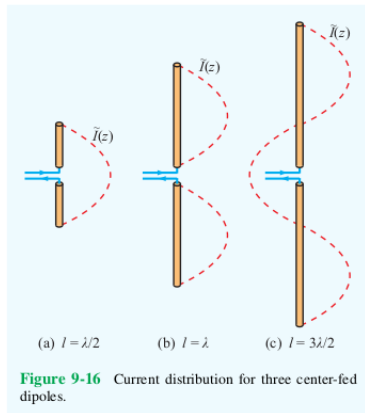
Radiation and Antennas (Continued)

Arbitrary-Length Dipole Antennas

For a **center-fed dipole**, the currents through the two halves are symmetrical and 0 at the ends $z = -\ell/2, 0, \ell/2$:

$$\tilde{I}(z) = I_0 \begin{cases} \sin(k(\frac{\ell}{2} - z)), & 0 \leq z \leq \frac{\ell}{2} \\ \sin(k(\frac{\ell}{2} + z)), & -\frac{\ell}{2} \leq z < 0 \end{cases} \quad (154)$$

where I_0 is the constant amplitude.



The **nonzero components of the EM field phasors** are

$$\tilde{E}_\theta = j60I_0 \left(\frac{e^{-jkR}}{R} \right) \left(\frac{\cos\left(\frac{k\ell}{2} \cos\theta\right) - \cos\left(\frac{k\ell}{2}\right)}{\sin\theta} \right), \quad \tilde{H}_\phi = \tilde{E}_\theta / \eta_0. \quad (155)$$

Radiation and Antennas (Continued)

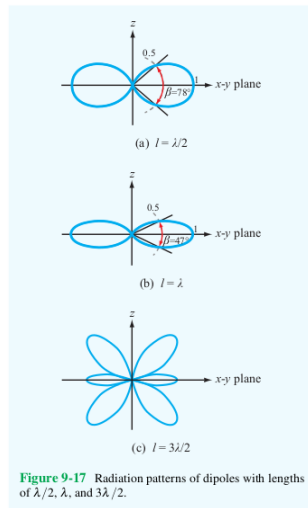
The **average power density** of the antenna is

$$S(R, \theta) = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi\ell}{\lambda} \cos\theta\right) - \cos\left(\frac{\pi\ell}{\lambda}\right)}{\sin\theta} \right)^2. \quad (156)$$

The **antenna patterns** when

$$\ell = \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2} \quad (157)$$

are shown in the adjacent figure. There is also a module accessible through the companion website that allows you to interactively explore the characteristics of this type of antenna.



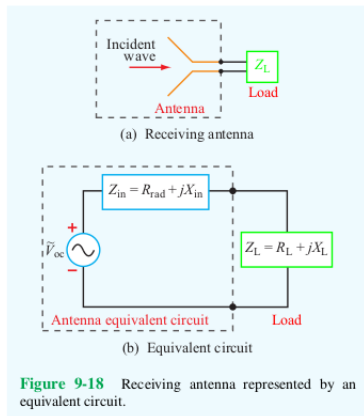
Receiving Antennas

Suppose

- S_i (W/m^2) is the power density of the incident EM wave
- and the antenna collects an **intercepted power** P_{int} (W).

Then the **effective area** (also called **effective aperture** or **receiving cross section**) of the antenna is defined as

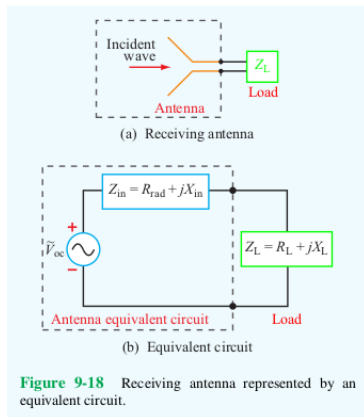
$$A_e = \frac{P_{\text{int}}}{S_i} \text{ (m}^2\text{)}. \quad (158)$$



Radiation and Antennas (Continued)

The **equivalent circuit** in the figure represents the transfer of power through the antenna to the load. Here,

- $\tilde{V}_{oc}$ is the **open-circuit voltage** that the incident wave induces at the antenna terminals,
- $Z_{in} = R_{in} + jX_{in}$ is the **input impedance** of the antenna,
- and $Z_L = R_L + jX_L$ is the **impedance of the load**.



Radiation and Antennas (Continued)

If

- $R_{\text{loss}} \ll R_{\text{rad}}$
- and Z_L and Z_{in} are chosen to make

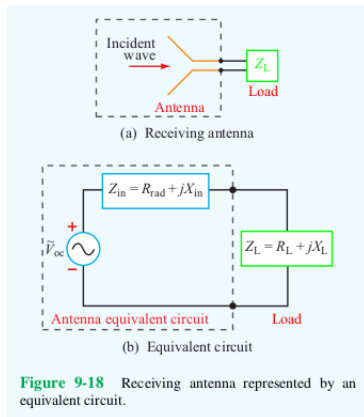
$$Z_L = Z_{\text{in}}^*, \quad (159)$$

then

- $Z_{\text{in}} = R_{\text{rad}} + jX_{\text{in}}$,
- the **overall impedance** is $2R_{\text{rad}}$,
- and the **maximum power**, which is **all of the intercepted power**,

$$P_L = P_{\text{int}} = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}} \quad (160)$$

is transferred to the load.



Radiation and Antennas (Continued)

Let $\tilde{E}_i$ be the component of the incident wave **parallel to the polarization direction** of the antenna. Then the **power density of the wave** is

$$S_i = \frac{|\tilde{E}_i|^2}{240\pi} \quad (161)$$

and the **effective area of the antenna** is

$$A_e = \frac{30\pi |\tilde{V}_{oc}|^2}{R_{rad} |\tilde{E}_i|^2}. \quad (162)$$

For any antenna,

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}. \quad (163)$$

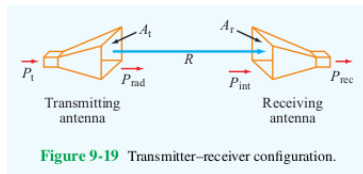
In particular, for a **Hertzian dipole**, $D = \frac{3}{2}$ (See Example 9-2.) and

$$A_e = \frac{3\lambda^2}{8\pi} \text{ (m}^2\text{)}. \quad (164)$$

Friis Transmission Formula

Suppose

- **two antennas** are oriented with the **peaks of their antenna patterns aligned**,
- one antenna **transmits**,
- the other antenna **receives**,
- and that the **transmission lines** connected to each antenna are **impedance matched** to the antennas.



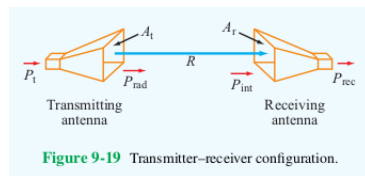
Radiation and Antennas (Continued)

For the **transmitting antenna**, let

- A_t be the **effective area**,
- ξ_t be the **radiation efficiency**,
- and P_t be the **power applied from the transmitter**.

For the **receiving antenna**, let

- A_r be the **effective area**,
- ξ_r be the **radiation efficiency**,
- and P_{rec} be the **power received at the receiver**.



Radiation and Antennas (Continued)

The **Friis transmission formula** determines the **power transfer ratio**

$$\frac{P_{\text{rec}}}{P_{\text{t}}} = \frac{\xi_{\text{t}}\xi_{\text{r}}A_{\text{t}}A_{\text{r}}}{\lambda^2 R^2} = G_{\text{t}}G_{\text{r}}\left(\frac{\lambda}{4\pi R}\right)^2. \quad (165)$$

More generally, if the **peaks of the antenna patterns are not aligned**,

$$\frac{P_{\text{rec}}}{P_{\text{t}}} = G_{\text{t}}G_{\text{r}}\left(\frac{\lambda}{4\pi R}\right)^2 F_{\text{t}}(\theta_{\text{t}}, \phi_{\text{t}})F_{\text{r}}(\theta_{\text{r}}, \phi_{\text{r}}). \quad (166)$$

Here,

- the angles $(\theta_{\text{t}}, \phi_{\text{t}})$ specify the direction of the receiving antenna relative to the antenna pattern of the transmitting antenna
- and the angles $(\theta_{\text{r}}, \phi_{\text{r}})$ specify the direction of the transmitting antenna relative to the antenna pattern of the receiving antenna.

Radiation and Antennas (Continued)

Aperture Antennas

An aperture antenna emits EM waves via the **electric field distributed across its aperture**.

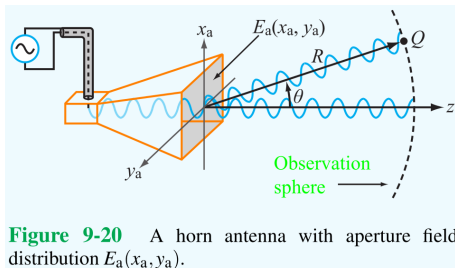


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

- Each point on the aperture can be thought of as an **isotropic antenna** radiating spherical waves.
- The waves from each point on the aperture are **superposed** to determine the **overall wave emitted**.

Radiation and Antennas (Continued)

Large-Aperture Antennas

A **scalar theory** due to Kirchhoff describes the **far-field behavior** of aperture antennas with principal dimensions that are **at least several wavelengths long**. For an observation point Q at a distance R from the aperture to be in the far field, R must satisfy the condition

$$R \geq 2d^2/\lambda, \quad (167)$$

where d is the **longest linear dimension** of the aperture.

The (scalar) electric field $E_a(x_a, y_a)$ at the aperture, called the

- **electric-field aperture distribution**
- or **illumination**,

is described using a Cartesian coordinate system (x_a, y_a, z) with its origin at the center of the aperture.

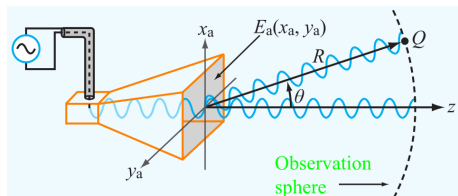


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

Radiation and Antennas (Continued)

For a rectangular aperture with length ℓ_x in the x_a direction and length ℓ_y in the y_a direction, the **phasor of the electric field** $\tilde{E}(R, \theta, \phi)$ at Q is

$$\tilde{E}(R, \theta, \phi) = \frac{j}{\lambda} \left(\frac{e^{-jkR}}{R} \right) \tilde{h}(\theta, \phi), \quad (168)$$

where (R, θ, ϕ) are the spherical (aperture) coordinates of Q ,

$$\tilde{h}(\theta, \phi) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \tilde{E}_a(x_a, y_a) e^{jk \sin \theta (x_a \cos \phi + y_a \sin \phi)} dx_a dy_a \quad (169)$$

is the **form factor**, and e^{-jkR}/R is the **spherical propagation factor**.

Assuming the unbounded medium has the **EM properties of free space**, the **power density** at Q is

$$S(R, \theta, \phi) = \frac{|\tilde{E}(R, \theta, \phi)|^2}{2\eta_0} = \frac{|\tilde{h}(\theta, \phi)|^2}{2\eta_0 \lambda^2 R^2} \quad (170)$$

and the polarization of $\tilde{E}(R, \theta, \phi)$ matches the polarization of $\tilde{E}_a(x_a, y_a)$.

Radiation and Antennas (Continued)

If the **illumination is uniform**, which means

$$\tilde{E}_a(x_a, y_a) = \begin{cases} E_0, & -\ell_x/2 \leq x_a \leq \ell_x/2, \quad -\ell_y/2 \leq y_a \leq \ell_y/2 \\ 0, & \text{otherwise} \end{cases}, \quad (171)$$

where E_0 is a **constant** and $\phi = 0$ so Q lies on the $+x$ axis, then the **form factor** is

$$\tilde{h}(\theta) = E_0 A_p \text{sinc}(\pi \ell_x \sin \theta / \lambda), \quad (172)$$

the **phasor of the electric field at Q** is

$$\tilde{E}(R, \theta, \phi) = \frac{j}{\lambda} \left(\frac{e^{-jkR}}{R} \right) \tilde{h}(\theta) = \frac{jE_0 A_p}{\lambda} \left(\frac{e^{-jkR}}{R} \right) \text{sinc}(\pi \ell_x \sin \theta / \lambda), \quad (173)$$

and the **power density at Q** is

$$S(R, \theta) = S_0 \text{sinc}^2(\pi \ell_x \sin \theta / \lambda), \quad (174)$$

where the **maximum power density** is

$$S_0 = S(R, 0) = \frac{E_0^2 A_p^2}{2\eta_0 \lambda^2 R^2}. \quad (175)$$

Radiation and Antennas (Continued)

Normalized radiation intensity:

$$F(\theta) = \text{sinc}^2(\pi \ell_x \sin \theta / \lambda) \quad (176)$$

Half-power beamwidths:

- x-z plane

$$\beta_{xz} \approx 0.88 \frac{\lambda}{\ell_x} \text{ (rad)} \quad (177)$$

- y-z plane

$$\beta_{yz} \approx 0.88 \frac{\lambda}{\ell_y}. \quad (178)$$

Uniform illumination yields the **narrowest beamwidth**.

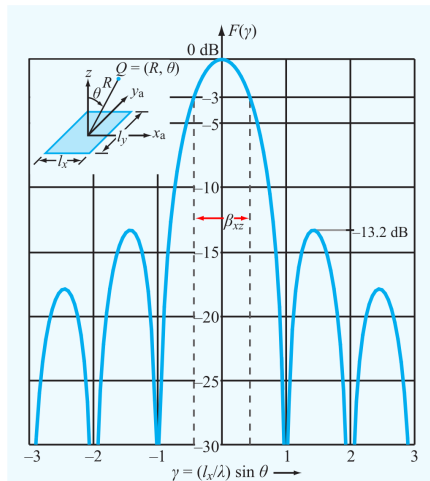


Figure 9-23 Normalized radiation pattern of a uniformly illuminated rectangular aperture in the x - z plane ($\phi = 0$).

Radiation and Antennas (Continued)

Directivity:

$$D \approx \frac{4\pi \ell_x \ell_y}{\lambda^2} = \frac{4\pi A_p}{\lambda^2}. \quad (179)$$

where A_p is the **physical area** of the aperture.

Effective area:

$$A_p \approx A_e \quad (180)$$

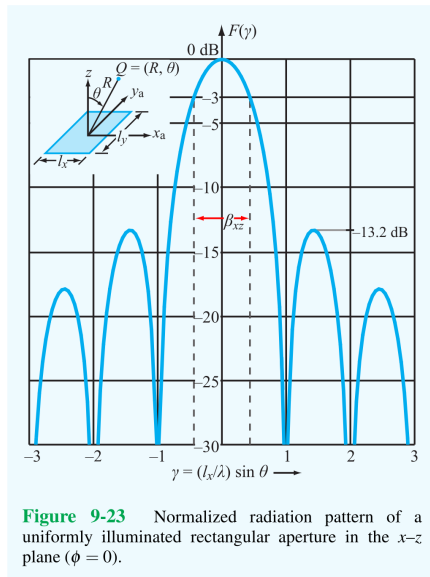


Figure 9-23 Normalized radiation pattern of a uniformly illuminated rectangular aperture in the x - z plane ($\phi = 0$).

Radiation and Antennas (Continued)

Antenna Arrays

When two or more antennas are combined into an **antenna array**, the direction and shape of the **overall antenna pattern can be controlled**. The antenna pattern of an array is controlled through the **amplitudes and relative phases of signals emitted** through the antennas.

Linear Antenna Arrays

We considered a **linear array of N identical transmitting antennas distributed along the z -axis**.

- All antennas **fed by a single oscillator** through a branching network.
- The **phase** and **amplitude** of the signal fed to each antenna is set by a **phase shifter** and an **amplifier** in the branch for the antenna.
- An antenna is an **element** of the array.
- Antenna/Element i is **located at** $(x_i, y_i, z_i) = (0, 0, id)$ for a **constant distance** d and $i = 0, \dots, N - 1$.

Radiation and Antennas (Continued)

The **phasor of the electric field intensity** in the far field for a single element of the array has the form

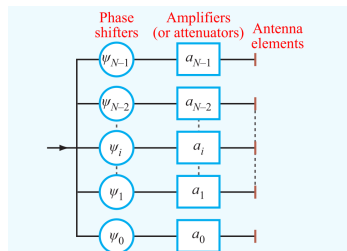
$$\tilde{E}_e(R, \theta, \phi) = \frac{e^{-jkR}}{R} \tilde{f}_e(\theta, \phi), \quad (181)$$

where

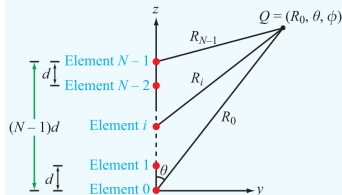
- e^{-jkR}/R is the **spherical propagation factor**
- and $\tilde{f}_e(\theta, \phi)$ describes the **dependence of $\tilde{E}_e$ on direction**.

The **average power density** due to $\tilde{E}_e$ is

$$S_e(R, \theta, \phi) = \frac{1}{2\eta_0 R^2} |\tilde{f}_e(\theta, \phi)|^2. \quad (182)$$



(a) Array elements with individual amplitude and phase control



(b) Array geometry relative to observation point

Radiation and Antennas (Continued)

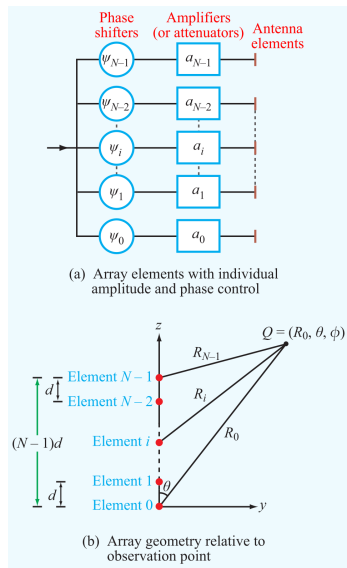
Suppose $Q = Q(R_0, \theta, \phi)$ is a point in the **far field of all of the antennas** and that R_i is the distance from antenna i to Q . This means

$$R_0 \geq \frac{2\ell^2}{\lambda} = \frac{2(N-1)^2 d^2}{\lambda}, \quad (183)$$

where

$$\ell = (N-1)d \quad (184)$$

is the **length of the array**.



Radiation and Antennas (Continued)

In this case, the **phasor of the electric field due to antenna i at Q** is

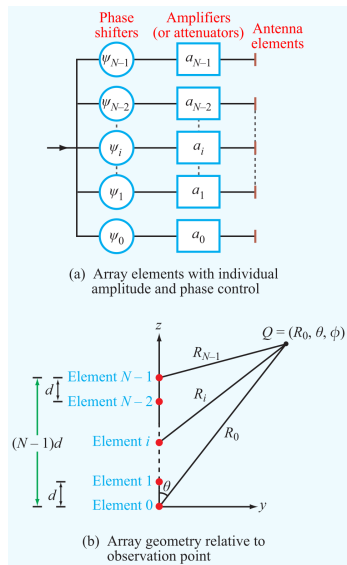
$$\tilde{E}_i(R_i, \theta, \phi) = A_i \frac{e^{-jkR_i}}{R_i} \tilde{f}_e(\theta, \phi), \quad (185)$$

where

$$A_i = a_i e^{j\psi_i} \quad (186)$$

is the **complex feeding coefficient** for antenna i . Here,

- the i^{th} phase shifter sets ψ_i
- and the i^{th} amplifier sets a_i .



Radiation and Antennas (Continued)

The **total electric field phasor** at Q is

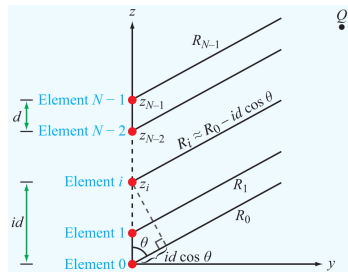
$$\tilde{E}(R_0, \theta, \phi) = \left(\sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \right) \tilde{f}_e(\theta, \phi). \quad (187)$$

Since Q is in the far field of all of the antennas,

$$\tilde{E}(R_0, \theta, \phi) = \tilde{f}_e(\theta, \phi) \left(\frac{e^{-jkR_0}}{R_0} \right) \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta}. \quad (188)$$

the **average power density** of the array is

$$\begin{aligned}
S(R_0, \theta, \phi) &= \frac{1}{2\eta_0} |\tilde{E}(R_0, \theta, \phi)|^2 \\
&= \frac{|\tilde{f}_e(\theta, \phi)|^2}{2\eta_0 R_0^2} \left| \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right|^2 \\
&= S_e(R_0, \theta, \phi) F_a(\theta). \quad (189)
\end{aligned}$$



Radiation and Antennas (Continued)

Here,

$$S_e(R_0, \theta, \phi) = \frac{|\tilde{f}_e(\theta, \phi)|^2}{2\eta_0 R_0^2} \quad (190)$$

is the **average power density for an individual element**, as before, and

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right|^2 = \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jkid \cos \theta} \right|^2 \quad (191)$$

is the **array factor**, which is the far-field radiation intensity of the array with isotropic radiators in place of the antennas.

The fact that

$$S(R_0, \theta, \phi) = S_e(R_0, \theta, \phi) F_a(\theta) \quad (192)$$

is called the **pattern multiplication principle**.

The sequence of magnitudes $a_0, a_1, \dots, a_{N-1}$ is the **array phase distribution** and the sequence of phases $\psi_0, \psi_1, \dots, \psi_{N-1}$ is the **array phase distribution**.

Radiation and Antennas (Continued)

We explored an example of **an array of two half-wave dipoles**.

N-Element Array with Equal Phase

When **all of the phases are equal**, i.e.

$$\psi_i = \psi_0 \text{ for all } i, \quad (193)$$

the array is a **broadside array** and the **array factor** is

$$F_a(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma} \right|^2, \quad (194)$$

where

$$\gamma = kd \cos \theta = \frac{2\pi d}{\lambda} \cos \theta. \quad (195)$$

Radiation and Antennas (Continued)

If **all of the amplitudes are also equal**, i.e.

$$a_i = a \in \mathbb{R} \text{ for all } i \text{ with } a > 0, \quad (196)$$

then the **array factor** is

$$F_a(\gamma) = a^2 \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}. \quad (197)$$

This has its maximum value

$$F_a(\gamma) = F_a(0) = a^2 N^2 = F_{a,\max} \quad (198)$$

when

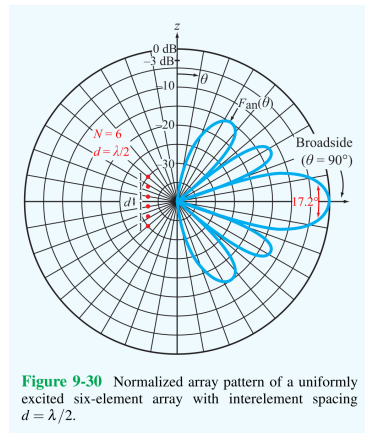
$$\theta = \pi/2 \quad (199)$$

and the **normalized array factor** is

$$F_{an} = \frac{\sin^2\left(\frac{N\pi d}{\lambda} \cos \theta\right)}{N^2 \sin^2\left(\frac{\pi d}{\lambda} \cos \theta\right)}. \quad (200)$$

Radiation and Antennas (Continued)

F_{an} could be the same as the antenna pattern but isn't always since the power density has an additional factor of $S_e(R_0, \theta, \phi)$, which is the power density of an individual element.



Radiation and Antennas (Continued)

Electronically Steering Antenna Arrays

A linear phase distribution

$$\psi_i = -i\delta \quad (201)$$

for some constant **incremental phase delay** δ modifies the **array factor** $F_a(\theta)$ and allows an array to be electronically steered. An array with this type of phase distribution is called a **linearly phased array**.

In this case, the **array factor** is

$$F_a(\theta) = F_a(\gamma') = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma'} \right|^2, \quad (202)$$

where

$$\gamma' = kd(\cos \theta - \cos \theta_0) = (2\pi d/\lambda)(\cos \theta - \cos \theta_0) \quad (203)$$

and θ_0 is the **scan angle** defined by

$$\delta = kd \cos \theta_0 = (2\pi d/\lambda) \cos \theta_0. \quad (204)$$

Radiation and Antennas (Continued)

The array factor $F_a(\gamma')$ has its maximum value when

$$\theta = \theta_0 \quad (205)$$

So, setting the **incremental phase delay in the linear phase distribution** to

$$\delta = \frac{2\pi d}{\lambda} \cos \theta_0 \quad (206)$$

steers the maximum power density to $\theta = \theta_0$.

Radiation and Antennas (Continued)

The **maximum power density of the array** occurs

- in the **broadside direction** $\theta_0 = \pi/2$ when $\delta = 0$,
- in the **end-fire direction** $\theta_0 = 0$ when $\delta = kd \cos 0 = kd = 2\pi d/\lambda$.

When the **amplitudes are equal**, the **normalized array factor** is

$$F_{an} = \frac{\sin^2\left(\frac{N\pi d}{\lambda}(\cos \theta - \cos \theta_0)\right)}{N^2 \sin^2\left(\frac{\pi d}{\lambda}(\cos \theta - \cos \theta_0)\right)} \quad (207)$$

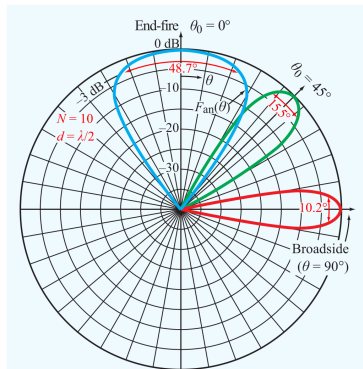


Figure 9-33 Normalized array pattern of a 10-element array with $\lambda/2$ spacing between adjacent elements. All elements are excited with equal amplitude. Through the application of linear phase across the array, the main beam can be steered from the broadside direction ($\theta_0 = 90^\circ$) to any scan angle θ_0 . Equiphase excitation corresponds to $\theta_0 = 90^\circ$.

Radiation and Antennas (Continued)

Array Feeding and Frequency Scanning

Frequency scanning is a practical approach to electronically controlling the direction of the maximum power density of an array. It means

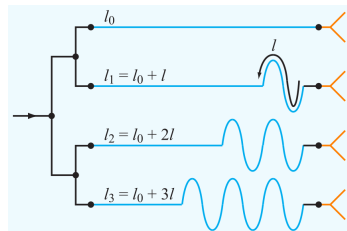
- using the **lengths of the transmission lines** that feed the elements of the array **to produce the phase delay**
- and **varying the frequency** of the signal sent to the antennas over a narrow frequency range centered on the **reference frequency** f_0 intended to be transmitted.

Suppose the **path length** for the transmission line connected to Element i is

$$\ell_i = \ell_0 + i\Delta\ell, \quad (208)$$

where

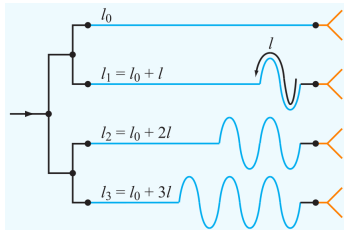
- ℓ_0 is the path length for Element 0
- and $\Delta\ell$ is a constant length increment.



Radiation and Antennas (Continued)

Choose the **length increment** to be a constant positive integer multiple of the wavelength λ_0 at f_0 , i.e.

$$\Delta\ell = n_0\lambda_0 = \frac{n_0 u_p}{f_0}. \quad (209)$$



Setting the phase $\psi_0 = 0$ and measuring the phase relative to ψ_0 ,

$$\psi_i(f) = -2in_0\pi \left(\frac{f}{f_0} \right). \quad (210)$$

- At the reference frequency $f = f_0$, $\theta = \pi/2$ and the array is broadside.
- At the frequency $f = f_0 + \Delta f$, $\theta = \theta_0$, where

$$\cos \theta_0 \approx \frac{n_0\lambda_0}{d} \left(\frac{\Delta f}{f_0} \right), \quad (211)$$

which is a good approximation if Δf is small enough.

Radiation and Antennas (Continued)

Note that

$$\cos \theta_0 \approx \frac{n_0 \lambda_0}{d} \left(\frac{\Delta f}{f_0} \right) \implies n_0 \approx \left(\frac{df_0}{\lambda_0 \Delta f} \right) \cos \theta_0. \quad (212)$$

So, there is a **tradeoff** between Δf and n_0 or, equivalently, $\Delta \ell$.

For a **desired scan angle** θ_0 and **fixed values for** d and f_0 :

- Choosing a **smaller** Δf means having to choose a **larger value for** n_0 and vice versa.

A **larger** n_0 means a **larger** $\Delta \ell$, which means **longer transmission lines**, which might be impractical.

Homework

Homework 4 is due tomorrow by 10 pm.

Exam 2 will be on Tuesday, June 9, from 1:30 – 3:20 pm.